Towards strong KPZ universality for a class of nonlinear SPDEs

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ABSTRACT. The purpose of this note is to bootstrap the weak KPZ universality from [44] for a class of nonlinear SPDEs to strong KPZ universality on polynomial time-scales. It is meant to be read as a natural follow-up (though somewhat longer than an appendix) to [44].

1. INTRODUCTION

The equations of interest here are the following *Ginzburg-Landau* SPDEs below, which we explain afterwards:

$$\partial_{\mathrm{t}}\mathbf{j}_{\mathrm{t},\mathrm{x}} = \partial_{\mathrm{x}}\mathscr{U}'(\partial_{\mathrm{x}}\mathbf{j}_{\mathrm{t},\mathrm{x}}) + \mathbf{F}[\mathscr{U}'(\partial_{\mathrm{x}}\mathbf{j}_{\mathrm{t},\mathrm{x}})] + \xi_{\mathrm{t},\mathrm{x}}. \tag{1.1}$$

Here, we take $(\mathrm{t},\mathrm{x}) \in [0,\infty)\times\mathbb{R}$, as well as functional parameters in the *potential* $\mathscr{U} : \mathbb{R} \to \mathbb{R}$ and the *nonlinearity* $\mathbf{F} : \mathbb{R} \to \mathbb{R}$. The term $\xi_{\mathrm{t},\mathrm{x}}$ is the Gaussian space-time white noise.

The Ginzburg-Landau SPDEs were popularized and systemized in statistical mechanics [22] as a robust model for a variety of phenomena, due to the flexibility in its functional parameters. They have also played important roles in the mathematical theory of hydrodynamic limits, especially; see [40] for a further discussion. In principle, however, the equation (1.1) is analytically ill-posed, since the noise ξ is too rough to be able to make sense of the RHS via purely analytic methods. Following standard protocol, we make sense of (1.1) as follows:

- We discretize (1.1) in space on a lattice of mesh N^{-1}, where $\mathsf{N} \to \infty$ is a scaling parameter. This produces an SDE that may be solved rigorously. (We could also consider continuous approximations, but to follow existing literature, in particular [44], we choose a discrete one.)
- We solve the discretized equation for each integer $\mathsf{N} \geqslant 1$.
- We show that the finite-N solutions converge as $\mathsf{N} \to \infty$, possibly with renormalization by N-dependent terms.

An important specialization of (1.1) is given by quadratic potential and nonlinearity. This produces the *Kardar-Parisi-Zhang* (KPZ) equation, whose well-posedness under the procedure above was shown in [18] (in the setting of the torus $\mathbb{R}/\mathbb{Z}$ instead of $\mathbb{R}$), with equivalent solution theories in [9, 17, 28], and extensions to the full-space in [35, 45]. The equation itself is given as follows:

$$\partial_{\mathrm{t}}\mathbf{h}_{\mathrm{t},\mathrm{x}} = \alpha\partial_{\mathrm{x}}^2\mathbf{h}_{\mathrm{t},\mathrm{x}} + \beta|\partial_{\mathrm{x}}\mathbf{h}_{\mathrm{t},\mathrm{x}}|^2 + \xi_{\mathrm{t},\mathrm{x}}, \tag{1.2}$$

Above, we take $\alpha > 0$ and $\beta \in \mathbb{R}$ to be fixed constants. Our main interest is for $\beta \neq 0$; otherwise, the solution to (1.2) is a Gaussian process. This SPDE was introduced in [24], which also proposes that it is a *universal* effective description of interface fluctuations. This is the *weak* KPZ *universality conjecture*, which, in this context, states that if the discretization above for (1.1) applies an additional *weak coupling constant* in front of $\mathbf{F}$ which vanishes as $\mathsf{N} \to \infty$ at an appropriate rate, then the limit obtained in the procedure above is the solution to (1.2) with α, β depending on $\mathscr{U}, \mathbf{F}$. (We clarify that the KPZ equation itself is invariant-in-law under this weak coupling scaling, hence its emergence as a scaling limit.) A (rigorous) resolution of this conjecture is the focus of an extensive line of research in [14, 16, 19, 20, 21, 26, 27, 42, 43], which restricted to the compact setting of $\mathbb{R}/\mathbb{Z}$, with a proof in the full-space coming in [44].

Another (outstanding) question concerning the asymptotics of (1.1) is the *strong* KPZ *universality conjecture*. In the current context, this essentially states that for some fixed constants $\mathrm{c}_1, \mathrm{c}_2 \neq 0$, the process

$$(\mathrm{t},\mathrm{x}) \mapsto \mathrm{c}_2\varepsilon^{\frac{1}{3}} \cdot (\mathbf{j}_{\varepsilon^{-1}\mathrm{t},\varepsilon^{-\frac{2}{3}}\mathrm{x}} - \mathrm{c}_1\varepsilon^{-1}\mathrm{t})$$

converges to the so-called KPZ *fixed point* that was recently constructed in [29] (with some notion of fundamental solution, i.e. the *directed landscape*, constructed in [13]). In the language of the previous discretization

scheme, it essentially says that the limit obtained in the bullet points above is the KPZ fixed point, if we apply an appropriate *strong coupling scaling* (depending on N) to the nonlinearity $\mathbf{F}$. For (1.1) in general, this remains open with, to our knowledge, minimal progress that we discuss further in Section 1.3.

In the specific case of (1.2), however, substantially more progress has been made, including the computation of the one-point distributions of the solution to (1.2) in the strong coupling limit for a few initial data [4, 8, 12]. We review this (and more results on strong universality) in Section 1.3. In particular, combining these with the universality result of [44] yields a strong-universality-type result for (1.1) under a double limit, given by first applying weak universality then strong universality scaling. More precisely, the estimates in [44] give

$$\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{|\mathrm{x}|\leqslant\mathrm{L}}|\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}-\mathbf{h}_{\mathrm{t},\mathrm{x}}|\leqslant \mathrm{e}^{\mathrm{CT}}\mathsf{N}^{-\gamma}, \tag{1.3}$$

where $\mathrm{C},\mathrm{L},\mathrm{T},\gamma>0$ are independent of N (with the first three being arbitrary), and where $\mathbf{j}^{\mathsf{N}}$ is the discretization of (1.1) at scale N. The estimate (1.3) thus yields strong universality up to *logarithmic* time-scales $\mathrm{T}\ll\gamma\log\mathsf{N}$. However, this is far from optimal (as we explain in Section 1.4 below). The motivation of this note is to therefore take a first step towards bridging the gap between what (1.3) gives and an optimal result.

1.1. **Our results.** The goal of this note is to bootstrap a logarithmic time-scale strong universality to *polynomial* time-scales. Precisely, we show there exists a constant $\mathrm{c}>0$ such that for any deterministic time $\mathrm{t}\leqslant\mathsf{N}^{\mathrm{c}}$,

$$\sup_{|\mathrm{x}|\leqslant\mathsf{N}^{\mathrm{c}}}|\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}-\mathbf{h}_{\mathrm{t},\mathrm{x}}|\leqslant\mathsf{N}^{-\mathrm{c}}, \tag{1.4}$$

where $\mathbf{j}^{\mathsf{N}}$ denotes the spatial-discretization of (1.1) at scale N with weak coupling scaling. This estimate will be obtained for a large class of potentials, for any polynomial nonlinearity, and for a large class of (non-equilibrium) initial data. To illustrate the result more clearly, we will combine (1.4) with known (strong coupling) asymptotics for the one-point distribution of the KPZ equation (e.g. in [4, 8, 12]) to get $\mathbf{j}^{\mathsf{N}}_{\mathrm{t},0}\approx \mathrm{c}_1\mathrm{t}+\mathrm{c}_2\boldsymbol{\chi}\mathrm{t}^{1/3}$ in law for $\mathrm{t}\leqslant\mathsf{N}^{\mathrm{c}}$ as $\mathsf{N}\to\infty$, where $\boldsymbol{\chi}$ is a random variable with an explicit law depending on the initial data of $\mathbf{j}^{\mathsf{N}}$. For instance, for "approximately" narrow-wedge initial data, we have $\boldsymbol{\chi}\sim\mathrm{TW}_2$, the law of the largest eigenvalue fluctuations in a GUE random matrix (in the infinite-dimensional limit). For Brownian initial data, we instead have $\boldsymbol{\chi}\sim\mathrm{BR}$, the Baik-Rains distribution (depending on GOE eigenvalue distribution formulas) in [5]. In general, however, the result intuitively states that strong universality results for the KPZ equation are automatically inherited by the more general family (1.1) up to polynomial time-scales, even process-level results.

We do not believe (1.4) to be optimal, but we do not expect it to be so far from optimal in the following sense. Strong universality predicts that $\mathbf{j}^{\mathsf{N}}\approx\mathbf{h}$ (in some sense) up to time-scales that diverge arbitrarily fast as $\mathsf{N}\to\infty$, which is well beyond the time-scales allowed in (1.4). But $\approx$ only needs to be in law, not path-wise, in principle. In fact, based on heuristics in Section 1.4 below, it is not clear to use if one can establish a path-wise comparison between (1.1) and (1.2) beyond $\mathrm{t}\ll\mathsf{N}^3$. Though there is still a gap between $\mathrm{t}\leqslant\mathsf{N}^{\mathrm{c}}$ with small $\mathrm{c}>0$ and $\mathrm{t}\ll\mathsf{N}^3$, this is a much smaller gap than the one between $\mathrm{t}\leqslant\mathsf{N}^{\mathrm{c}}$ and $\mathrm{t}\to\infty$ arbitrarily fast in N.

1.2. **Technical aspects.** Key to our bootstrapping from log-scales to polynomial-scales is the fact that the KPZ equation has a linearization which satisfies the parabolic comparison principle. Heuristically:

- Let $\boldsymbol{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{KPZ}}[\cdot]$ be the map which solves the KPZ equation to time t with time-s data given by the input $\cdot$, where $0\leqslant\mathrm{s}\leqslant\mathrm{t}$. Let $\boldsymbol{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{GL}}[\cdot]$ be the same but for the evolution of $\mathbf{j}^{\mathsf{N}}$. By the flow property, we have
$$\boldsymbol{\Phi}^{0,\mathrm{t}}_{\mathrm{KPZ}}=\boldsymbol{\Phi}^{1,\mathrm{t}}_{\mathrm{KPZ}}\circ\boldsymbol{\Phi}^{0,1}_{\mathrm{KPZ}}=\boldsymbol{\Phi}^{1,\mathrm{t}}_{\mathrm{KPZ}}\circ\boldsymbol{\Phi}^{0,1}_{\mathrm{GL}}+\Big(\boldsymbol{\Phi}^{1,\mathrm{t}}_{\mathrm{KPZ}}\circ\boldsymbol{\Phi}^{0,1}_{\mathrm{KPZ}}-\boldsymbol{\Phi}^{1,\mathrm{t}}_{\mathrm{KPZ}}\circ\boldsymbol{\Phi}^{0,1}_{\mathrm{GL}}\Big).$$
In words, we modify the KPZ equation flow by replacing the $[0,1]$-piece by the flow on $[0,1]$ for $\mathbf{j}^{\mathsf{N}}$.
- The input (1.3) from [44] yields $|\boldsymbol{\Phi}^{0,1}_{\mathrm{KPZ}}-\boldsymbol{\Phi}^{0,1}_{\mathrm{GL}}|\leqslant\mathrm{C}_0\cdot\mathsf{N}^{-\gamma}$ (in some sense to be made precise) for some finite constant $\mathrm{C}_0>0$.
- By the aforementioned parabolic maximum principle (for the linearization of the KPZ equation), and a host of technical gymnastics to account for the spatially-compact nature of (1.3) and the global nature of comparison principles, we know that the size of $|\boldsymbol{\Phi}^{0,1}_{\mathrm{KPZ}}-\boldsymbol{\Phi}^{0,1}_{\mathrm{GL}}|$ is monotonically non-increasing under the $\boldsymbol{\Phi}^{1,\mathrm{t}}_{\mathrm{KPZ}}$-flow (up to ultimately negligible errors). Thus, the term in parentheses is $\leqslant\mathrm{C}_0\cdot\mathsf{N}^{-\gamma}$ globally in time.
- We continue inductively, replacing the $[1,2]$-piece in $\boldsymbol{\Phi}^{1,\mathrm{t}}_{\mathrm{KPZ}}$ by $\boldsymbol{\Phi}^{1,2}_{\mathrm{GL}}$ with an additive cost of $\leqslant\mathrm{C}_0\cdot\mathsf{N}^{-\gamma}$. If we do this $\leqslant\mathsf{N}^{\mathrm{c}}$-many times, then the cost of $\leqslant\mathrm{C}_0\cdot\mathsf{N}^{\mathrm{c}}\mathsf{N}^{-\gamma}\ll 1$, so we win.

As alluded to already, there are many details which must be addressed to make this strategy work. For example, we must impose assumptions on initial data that are stable (and do not blow up) as we evolve under the $\mathbf{\Phi}_{\mathrm{GL}}^{0,\mathrm{t}}$-flow. For this reason, it will be convenient to work with invariant measure initial data for the $\mathbf{\Phi}_{\mathrm{GL}}^{0,\mathrm{t}}$-flow, with estimates that are quantitative enough to extend to non-equilibrium data after. We use spatially-global weighted estimates for the $\mathbf{\Phi}_{\mathrm{KPZ}}^{0,\mathrm{t}}$ and $\mathbf{\Phi}_{\mathrm{KPZ}}^{0,\mathrm{t}}$ flows which are locally uniform in time as well, as to localize the parabolic comparison principle used above (and make it compatible with the local nature of (1.3)).

1.3. **Background.** To our knowledge, the first strong universality result for (1.1) comes from [4], which showed convergence of one-point distributions for the narrow-wedge solution of the KPZ equation (defined by an initial data that is formally given by $\alpha\beta^{-1} \cdot \log \delta_0$) to the TW_2 law governing the fluctuations of the largest eigenvalue in a GUE matrix (in the infinite-dimensional limit). At the same time, non-rigorous versions of these results were obtained in the physics literature [37, 38, 39]. We remark that these results hold only for the specialization (1.2). Extensions to other initial data with other special distributions were shown in [8, 12].

Afterwards, a lift of the narrow-wedge solution to KPZ, viewed as a process in space evaluated at a fixed time, was constructed and studied in great depth in [11]. Such lift, which is now referred to as the KPZ *line ensemble*, is a collection of curves containing the spatial process of the KPZ equation, and in [11], it was predicted to converge under 1:2:3 scaling to the *Airy line ensemble* that was studied extensively in [10]. A proof was given in [3, 41].

For (1.1) in general, see [31, 32, 33, 34], which shows tail estimates for the one-point distribution that are stable under the 1:2:3 KPZ, for a general collection of potentials $\mathscr{U}$ (and linear $\mathbf{F}$). These works also include extensions of the method to other (non-integrable) models in the KPZ universality class.

As for a complete strong universality result (namely, convergence as a process to the KPZ fixed point), we cite the works [1, 2, 29], which show such a convergence for interfaces associated to integrable models, including TASEP, ASEP, and the stochastic six-vertex model, which can all be viewed as microscopic versions of (1.1).

Finally, we recognize that this is far from a complete account of results related to strong KPZ universality; we have focused on results that are more closely related to universality for the SPDEs (1.1) of main interest herein.

1.4. **Heuristics.** The (path-wise) comparison between $\mathbf{j}^{\mathsf{N}}$ and $\mathbf{h}$ is ultimately determined by fluctuations of $\mathscr{U}, \mathbf{F}$ around quadratic choices of these functional parameters. Analysis in [44] suggests that these fluctuations behave like space-time noises and should thus be of order $\mathsf{N}^{-1/2}\mathrm{t}^{1/2}$. (The reason for $\mathsf{N}^{-1/2}$ is given in after Definition 4.2 in [44]; it essentially reflects square-root CLT cancellations upon averaging at scale N.) It is also known, e.g. by work [4], that the fluctuations of $\mathbf{h}$ behave like $\mathrm{t}^{1/3}$ in time. This leaves us with $\mathsf{N}^{-1/2}\mathrm{t}^{1/2} \ll \mathrm{t}^{1/3}$, i.e. $\mathrm{t} \ll \mathsf{N}^3$, as mentioned above. However, the difficulty in making an argument along these lines precise is that this estimate $\mathsf{N}^{-1/2}\mathrm{t}^{1/2}$ is always fed back into a Gronwall argument, hence the exponential factor in (1.3).

Let us give a second heuristic which yields the same condition $\mathrm{t} \ll \mathsf{N}^3$ through analogy with another model in the KPZ universality class. The $\mathbf{j}^{\mathsf{N}}$ dynamics are driven by the restriction of a space-time white noise to a spatial lattice of mesh N^{-1}. Thus, this discrepancy between this noise and the continuum space-time white noise driving (1.2) also lives at scale N^{-1}. In parallel with results of [15] (for a model of "Brownian last passage percolation"), pathwise decorrelation between $\mathbf{j}^{\mathsf{N}}$ and (1.2) should emerge if the noise discrepancies dominate the third-root of the time-scale, namely $\mathsf{N}^{-1} \gg \mathrm{t}^{-1/3}$, or equivalently $\mathrm{t} \gg \mathsf{N}^3$.

1.5. AI **statement.** This project began by asking an LLM[1] to double-check our understanding of how the weak universality result of [44] provides strong universality up to logarithmic scales (via the estimate (1.3)). After this, we consulted an LLM[2] on bootstrapping to polynomial time-scales. We proposed a scheme based on comparison principles (motivated by prior work [42] of the author). To be precise about what AI contributed mathematically:

- The LLM proposed the shifting idea in the proof of Proposition 3.4 (only a vague sentence or two was given; we turned this into a full argument without the LLM assistance to double check our understanding of its claims).
- The LLM proposed the idea of modifying the construction in [44] of an approximate narrow-wedge initial data. That is, it proposed to allow the initial data depend on the pre-determined time-scale t_{N} of interest in

[1]Claude Opus 5.5, with access granted via the Claude-for-scientists program.
[2]Claude Opus 5.5, in a different chat to double our reasoning.

Corollary 2.9. The structure of the initial data is the same as in [44], but the idea to allow it to depend on a pre-determined time-scale at which we evaluate the KPZ equation is clever and unlocks Corollary 2.9 (though it has no bearing on Theorem 2.6, Corollary 2.7, or Theorem 2.8).

In addition to the above, we also used the LLM in the following ways for the writing of this note:

- We consulted the LLM on how to avoid the Cole-Hopf transform for estimates on the full-space KPZ equation. We proposed ideas based on [35, 45], though this did not succeed, since the topologies needed in this paper are not compatible with those used in [35, 45]. (This is technical, though, and it is *not* a comment on the strength of [35, 45]; presumably, the estimates in [35, 45] may be carefully tracked to produce the results needed here.)
- We used the LLM to scan [11] for estimates on the narrow-wedge solution of the KPZ equation which are stable under 1:2:3 scaling. We found this to be very helpful in our workflow and emphasizing to us the value of [11].
- Finally, the LLM was used to proofread and double-check the arguments in this paper.

The author assumes responsibility for all the content in this paper.

2. Main results

We start with introducing the model of interest in this paper; this is borrowed from [44], but with an additional rescaling in space (to macroscopic coordinates). First, we introduce some important notation:

- The parameter $\mathsf{N} \geqslant 1$ is an integer that we will take to ∞; we call this the large-N limit.
- We will set $\mathbb{X}_{\mathsf{N}} := \mathsf{N}^{-1}\mathbb{Z}$ as the discretized spatial-domain at scale N.
- Elements of $\mathbb{R}^{\mathbb{X}_{\mathsf{N}}}$ are denoted by $\boldsymbol{\phi} := (\boldsymbol{\phi}_{\mathrm{x}})_{\mathrm{x}\in\mathbb{X}_{\mathsf{N}}}$. Fix $\mathrm{w} \in \mathbb{X}_{\mathsf{N}}$; we define the corresponding shift $\tau_{\mathrm{w}} : \mathbb{R}^{\mathbb{X}_{\mathsf{N}}} \to \mathbb{R}^{\mathbb{X}_{\mathsf{N}}}$ via $\tau_{\mathrm{w}}\boldsymbol{\phi} := (\boldsymbol{\phi}_{\mathrm{x}+\mathrm{w}})_{\mathrm{x}\in\mathbb{X}_{\mathsf{N}}}$ for any $\boldsymbol{\phi} \in \mathbb{R}^{\mathbb{X}_{\mathsf{N}}}$, which, in words, re-centers the configuration $\boldsymbol{\phi}$ around $\mathrm{w} \in \mathbb{X}_{\mathsf{N}}$.
- Finally, for any $\mathrm{a}, \mathrm{b} \in \mathbb{R}$, we set $[\![\mathrm{a}, \mathrm{b}]\!] := [\mathrm{a}, \mathrm{b}] \cap \mathbb{Z}$.

We now present the following space-discretization (on $\mathbb{X}_{\mathsf{N}}$) of (1.1), with notation explained after:

$$\mathrm{d}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{1}(\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}])\mathrm{d}\mathrm{t} + \mathsf{N}\mathbf{F}[\tau_{\mathrm{x}}\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathrm{t} + \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}} - \mathscr{R}_{\lambda}\mathrm{d}\mathrm{t}. \tag{2.1}$$

Above, we have $(\mathrm{t}, \mathrm{x}) \in [0, \infty) \times \mathbb{X}_{\mathsf{N}}$. Moreover, as in [44], the factor of $\sqrt{2}$ is only a convention, and we used the following notation:

- For any $\mathfrak{l} \in \mathbb{Z}$, we define the discrete gradient $\nabla^{\mathbf{X}}_{\mathfrak{l}}$ on $\mathbb{X}_{\mathsf{N}}$ as follows; for any function $\mathsf{f} : \mathbb{X}_{\mathsf{N}} \to \mathbb{R}$, we define its gradient as another function $\nabla^{\mathbf{X}}_{\mathfrak{l}}\mathsf{f}_{\mathrm{x}} := \mathsf{f}_{\mathrm{x}+\mathsf{N}^{-1}\mathfrak{l}} - \mathsf{f}_{\mathrm{x}}$ on $\mathbb{X}_{\mathsf{N}}$.
- We use the field-theoretic notation $\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} := -\mathsf{N}^{1/2}\nabla^{\mathbf{X}}_{-1}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}$.
- Above, the function $\mathscr{U} : \mathbb{R} \to \mathbb{R}$ is the potential, which we impose assumptions on in Assumption 2.2 below.
- The processes $\mathrm{t} \mapsto \mathbf{b}_{\mathrm{t},\mathrm{x}}$ are independent standard Brownian motions.
- The nonlinearity $\mathbf{F}$ is defined as the following polynomial-type function:

$$\begin{aligned}\mathbf{F}[\boldsymbol{\phi}] := \mathbf{F}_2[\boldsymbol{\phi}] + \mathbf{F}_{>2}[\boldsymbol{\phi}] := \tfrac{1}{3}\beta_2(\mathscr{U}'[\boldsymbol{\phi}_{\frac{2}{\mathsf{N}}}]\mathscr{U}'[\boldsymbol{\phi}_{\frac{1}{\mathsf{N}}}] + \mathscr{U}'[\boldsymbol{\phi}_{\frac{1}{\mathsf{N}}}]\mathscr{U}'[\boldsymbol{\phi}_{0}] + \mathscr{U}'[\boldsymbol{\phi}_{0}]\mathscr{U}'[\boldsymbol{\phi}_{-\frac{1}{\mathsf{N}}}]) \\ + \sum_{\mathrm{d}\in[\![3,\deg]\!]}\beta_{\mathrm{d}}\Big(\prod_{\ell\in[\![1,\mathrm{d}]\!]}\mathscr{U}'[\boldsymbol{\phi}_{\frac{\ell}{\mathsf{N}}}] + \ldots + \prod_{\ell\in[\![1,\mathrm{d}]\!]}\mathscr{U}'[\boldsymbol{\phi}_{-\frac{\ell}{\mathsf{N}}+\frac{1}{\mathsf{N}}}]\Big).\end{aligned} \tag{2.2}$$

- The constant $\mathscr{R}_{\lambda}$ here is defined later in (2.7), after we have introduced the necessary notation. The reason for this $\mathscr{R}_{\lambda}$-term in (2.1) is to account for a renormalization counter-term. (The paper [44] does not introduce this counter-term into the dyamics of $\mathbf{j}^{\mathsf{N}}$, but all convergence results stated therein hold for $\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathscr{R}_{\lambda}\mathrm{t}$. Finally, we clarify that $\mathscr{R}_{\lambda}$ is independent of N, as demonstrated by (2.6)-(2.7).)

The discretization (2.1)-(2.2) here has the special (though natural) property in that it has the following probability measures as invariant ones. (This is shown in Section A.3 of [44].)

Definition 2.1. Fix any $\sigma \in \mathbb{R}$. We let $\mathbb{P}^{\sigma}$ be the following product measure on $\mathbb{R}^{\mathbb{X}_{\mathsf{N}}}$:

$$\mathbb{P}^{\sigma}[\mathrm{d}\boldsymbol{\phi}] := \prod_{\mathrm{x}\in\mathbb{X}_{\mathsf{N}}}\boldsymbol{\mathcal{Z}}^{-1}_{\sigma,\mathscr{U}}\mathbf{Exp}(-\mathscr{U}[\boldsymbol{\phi}_{\mathrm{x}}] + \upsilon_{\sigma}\boldsymbol{\phi}_{\mathrm{x}})\mathrm{d}\boldsymbol{\phi}_{\mathrm{x}}.$$

The constant $\mathcal{Z}_{\sigma,\mathscr{U}} > 0$ is chosen so that $\mathbb{P}^\sigma$ is a probability measure, and $\upsilon_\sigma \in \mathbb{R}$ is chosen so that $\mathbb{E}^\sigma \boldsymbol{\phi}_{\mathrm{x}} = \sigma$ for all x, where $\mathbb{E}^\sigma$ denotes the expectation with respect to $\mathbb{P}^\sigma$. We will refer to $\mathbb{P}^\sigma$ as the *grand-canonical ensemble* or *grand-canonical measure* of density $\sigma \in \mathbb{R}$.

To ensure that the previous grand-canonical ensembles are well-defined, we impose the following assumption on the potential $\mathscr{U}$ (which is also key to the analysis in [44] on universality of the KPZ equation for (2.1)).

Assumption 2.2. *Suppose $\mathscr{U} = \mathscr{U}_1 + \mathscr{U}_2$, where $\mathscr{U}_1 \in \mathscr{C}^\infty(\mathbb{R},\mathbb{R})$ satisfies $\mathscr{U}_1''(\mathrm{a}) \in [\boldsymbol{\Lambda}^{-1}, \boldsymbol{\Lambda}]$ for each $\mathrm{a} \in \mathbb{R}$ and some fixed $\boldsymbol{\Lambda} \geqslant 1$, and where $\mathscr{U}_2 \in \mathscr{C}^\infty(\mathbb{R},\mathbb{R})$ satisfies $\|\mathscr{U}_2\|_{\mathscr{C}^2(\mathbb{R})} < \infty$. We emphasize that all functions here are independent of N. Finally, for convenience (in the analysis of [44]), we assume that $\upsilon_0 = 0$ and $\mathbb{E}^0|\boldsymbol{\phi}_0|^2 = 1$.*

We clarify that the assumption $\upsilon_0 = 0$ is without loss of generality upon re-defining $\mathscr{U}_1$ by a linear shift. The other assumption $\mathbb{E}^0|\boldsymbol{\phi}_0|^2 = 1$ is for convenience only, since it fixes α in the limiting SPDE to be 1. In particular, this assumption is ultimately also without loss of generality after rescaling (1.2) as illustrated in [36].

Now, we present assumptions on the initial data for (2.1). The first will be adopted throughout this paper.

Assumption 2.3. *We have $\mathbf{j}^{\mathsf{N}}_{0,0} = 0$ deterministically.*

Assumption 2.3 is harmless, because global (space-time-independent) additive shifts of $\mathbf{j}^{\mathsf{N}}$ commute with the solution map corresponding to (2.1); this follows because the RHS of (2.1) depends only on the gradient of $\mathbf{j}^{\mathsf{N}}$.

We now give two separate assumptions for the rest of the initial profile of $\mathbf{j}^{\mathsf{N}}$. In words, Assumption 2.4 asserts equilibrium data, and Assumption 2.5 allows a much larger class of non-equilibrium initial data. We clarify these afterwards.

Assumption 2.4. *We have $\boldsymbol{\phi}_{0,\cdot} \sim \mathbb{P}^0$.*

Assumption 2.5. *Suppose that $\mathbb{P}^{\mathrm{init}}$ is the law of $\boldsymbol{\phi}_{0,\cdot}$, and set $\mathfrak{P}$ such that $\mathrm{d}\mathbb{P}^{\mathrm{init}} = \mathfrak{P}\mathrm{d}\mathbb{P}^0$; that is, $\mathfrak{P}$ is the density of the law of $\boldsymbol{\phi}_{0,\cdot}$ with respect to $\mathbb{P}^0$. For some (small but N-independent) $\gamma_{\mathrm{data}} > 0$, we have*

$$\|\mathfrak{P}\|^2_{\mathrm{L}^2(\mathbb{P}^0)} \leqslant \mathsf{N}^{\gamma_{\mathrm{data}}}. \tag{2.3}$$

It is not so important that we adopt L^2 on the LHS of (2.3) in particular; any L^p-norm for $p > 1$ fixed is okay (if we adjust $\gamma > 0$). Indeed, we will first establish results under the equilibrium setting of Assumption 2.4 with quantitative high probability estimates, which will then transfer to non-equilibrium data in Assumption 2.5. We clarify that while (2.3) is perturbative in the sense of requiring $\gamma_{\mathrm{data}} > 0$ to be small, it still allows us to consider a large class of non-equilibrium data including the narrow-wedge (plus vanishingly small fluctuations), exactly because the RHS of (2.3) is allowed to diverge in the large-N limit. This point is explained in detail in [44]. At the same time, there is a gap between (2.3) and the non-equilibrium assumption in [44] (which is *not* perturbative).

2.1. **Main results.** As alluded to earlier, we will first present the result for equilibrium initial data, from which we transfer to data satisfying the a priori estimate (2.3). Before we state either result, we define

$$\alpha := \partial_\sigma \mathbb{E}^\sigma \mathscr{U}'[\boldsymbol{\phi}_0]|_{\sigma=0}, \tag{2.4}$$

$$\beta := \tfrac{1}{2}\partial_\sigma^2 \mathbb{E}^\sigma \mathbf{F}[\boldsymbol{\phi}]|_{\sigma=0}, \tag{2.5}$$

$$\lambda := \alpha^{-1}\beta, \tag{2.6}$$

$$\mathscr{R}_\lambda := \tfrac{1}{12}\lambda^3 \mathbb{E}^0(\mathscr{U}'[\boldsymbol{\phi}_0]\boldsymbol{\phi}_0^3) - \tfrac{1}{6}\beta_2\lambda^2. \tag{2.7}$$

As shown in Section A.2 of [44], we have $\beta \neq 0$ if and only if $\beta_2 \neq 0$. Thus, we will assume that $\beta_2 \neq 0$.

Finally, before we proceed, we adopt the standard big-O notation. That is, we write $\mathrm{a} = \mathrm{O}(\mathrm{b})$ (or equivalently, $\mathrm{a} \lesssim \mathrm{b}$) if $|\mathrm{a}| \leqslant \mathrm{C}|\mathrm{b}|$ for some constant $\mathrm{C} > 0$. Parameters on which C depends will be listed as subscripts in the O or $\lesssim$ notation, with no subscripts indicating that $\mathrm{C} > 0$ is a fixed universal constant. Moreover, we will write $\mathrm{a} = \mathrm{o}(\mathrm{b})$ (or equivalently, $\mathrm{a} \ll \mathrm{b}$), if $\frac{\mathrm{a}}{\mathrm{b}} \to 0$ as $\mathsf{N} \to \infty$.

Theorem 2.6. *Suppose that Assumptions 2.3 and 2.4 hold. There exists a coupling between $\mathbf{j}^{\mathsf{N}}$ and the solution $\mathbf{h}$ to (1.2) with α, β from (2.4)-(2.5) and initial data given by a standard two-sided Brownian motion so that for some $\mathrm{c} > 0$ independent of N and for any deterministic $\mathrm{t} \in [0, \mathrm{O}(\mathsf{N}^{\mathrm{c}})]$, with probability $1 - \mathrm{O}(\mathsf{N}^{-\mathrm{c}})$, we have*

$$\sup_{|\mathrm{x}| \leqslant \mathrm{O}(\mathsf{N}^{\mathrm{c}})} |\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{h}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\mathrm{c}}. \tag{2.8}$$

We clarify that by standard two-sided Brownian motion, we mean a Gaussian process $\mathbf{w}$ *on* $\mathbb{R}$ *with independent increments such that* $\mathbf{w}_{\mathrm{x}} - \mathbf{w}_{\mathrm{y}} \sim \mathcal{N}(0, |\mathrm{x} - \mathrm{y}|)$ *and* $\mathbf{w}_0 = 0$.

Let us clarify that the coupling referred to in Theorem 2.6 has a static and dynamic component to it. First, we couple the initial data of $\mathbf{j}^{\mathsf{N}}$, which is a random walk with increments distributed according to $\mathbb{P}^0$ by Assumptions 2.3 and 2.4, to a two-sided Brownian motion. This is ultimately done through the KMT coupling [25]. Second, we couple the Brownian motions in (2.1) to a space-time white noise ξ such that as a space-time field, said Brownian motions converge to the space-time white noise in a path-wise sense.

Observe that the statement of Theorem 2.6 gets weaker as $\mathrm{c} > 0$ gets smaller. Moreover, recall that $\mathbb{E}^0|\phi_0|^2 = 1$ actually gives $\alpha = 1$ (though, again, we keep this α-dependence to more directly parallel [44]), so that two-sided Brownian motions are, indeed, invariant in law under the KPZ equation flow.

As an immediate consequence of Theorem 2.6 and results [8] for long-time statistics of KPZ with Brownian initial data, we deduce the following one-point result for $\mathbf{j}^{\mathsf{N}}$ up to polynomial time-scales. We emphasize that a process-level statement would be available immediately after showing the same for the KPZ equation itself.

Corollary 2.7. *Adopt the setting of Theorem 2.6. There exist deterministic constants* $\mathrm{c}_1, \mathrm{c}_2 \in \mathbb{R}$ *with* $\mathrm{c}_2 \neq 0$ *and* $\mathrm{c}_3 > 0$ *so that for any deterministic* $1 \ll \mathsf{t}_{\mathsf{N}} \leqslant \mathsf{N}^{\mathrm{c}_3}$*, we have*

$$(\mathrm{c}_2 \mathsf{t}_{\mathsf{N}}^{\frac{1}{3}})^{-1} \cdot (\mathbf{j}^{\mathsf{N}}_{\mathsf{t}_{\mathsf{N}},0} - \mathrm{c}_1 \mathsf{t}_{\mathsf{N}}) \Rightarrow \mathrm{BR},$$

where the convergence is in distribution as $\mathsf{N} \to \infty$*, and where* BR *is the Baik-Rains distribution (see* [5]*).*

We now move to non-equilibrium initial data.

Theorem 2.8. *Suppose that Assumptions 2.3 and 2.5 (with small enough but fixed* $\gamma_{\mathrm{data}} > 0$*) hold. There exists a coupling between* $\mathbf{j}^{\mathsf{N}}$ *and the solution to* (1.2) *with* α, β *in* (2.4)-(2.5) *and initial data given by that of* $\mathbf{j}^{\mathsf{N}}$ *so that:*

- *For some constant* $\mathrm{c} > 0$ *independent of* N *and for any deterministic* $\mathrm{t} \in [0, \mathrm{O}(\mathsf{N}^{\mathrm{c}})]$*, we have the estimate* (2.8) *with probability* $1 - \mathrm{O}(\mathsf{N}^{-\mathrm{c}})$.

(To clarify, the initial data for (1.2) *is the extension of that of* $\mathbf{j}^{\mathsf{N}}$ *from* $\mathbb{X}_{\mathsf{N}}$ *to* $\mathbb{R}$ *via linear interpolation.)*

Examples of special initial data that satisfy Assumptions 2.3 and 2.5 include vanishingly small perturbations of narrow-wedge initial data, whose long-time statistics are distinguished (random matrix) statistics. The result below extends Corollary 2.7 to this case.

Corollary 2.9. *There exist deterministic constants* $\mathrm{c}_1, \mathrm{c}_2 \in \mathbb{R}$ *with* $\mathrm{c}_2 \neq 0$ *and* $\mathrm{c}_3 > 0$ *so that the following holds. Consider any* $1 \ll \mathsf{t}_{\mathsf{N}} \ll \mathsf{N}^{\mathrm{c}_3}$*. There are initial data satisfying Assumptions 2.3 and 2.5 so that*

$$(\mathrm{c}_2 \mathsf{t}_{\mathsf{N}}^{\frac{1}{3}})^{-1} \cdot (\mathbf{j}^{\mathsf{N}}_{\mathsf{t}_{\mathsf{N}},0} - \mathrm{c}_1 \mathsf{t}_{\mathsf{N}}) \Rightarrow \mathrm{TW}_2.$$

The convergence is in distribution as $\mathsf{N} \to \infty$*, and* TW_2 *is the Tracy-Widom* GUE *distribution (see* [4]*).*

The proof of Corollary 2.9 resembles that of Corollary 2.7, but it requires a construction of the desired initial data as well, so we defer it to Section 5.

2.2. **Organization of this paper.** In Section 3, we obtain the inputs from universality of the KPZ equation that we then bootstrap to strong universality on polynomial time-scales. In Section 4, we will perform this bootstrap to show Theorem 2.6 and Corollary 2.7. In Section 5, we use Theorem 2.6 to show Theorem 2.8, and we perform additional gymnastics to deduce Corollary 2.9 from Theorem 2.8.

3. Inputs from universality of the KPZ equation

In this section, we will gather and establish a priori estimates concerning the convergence of (2.1) to the KPZ equation on finite time-scales. The results of this section are essentially obtained by refining the analysis within [44] in ways that were not needed for convergence to the KPZ equation. We emphasize that Assumptions 2.2, 2.3, 2.4, and 2.5 all imply the assumptions in [44], and thus we can take from [44] anything therein.

Before we start, we declare the following assumption to be used throughout this section:

Suppose that Assumptions 2.3 and 2.4 hold.

Because [44] is in the language of exponentials (i.e. Cole-Hopf transforms) of (1.2) and (2.1), we will introduce the following notation. Recalling (2.6)-(2.7), we set (for $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times \mathbb{X}_{\mathsf{N}}$)

$$\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} := \mathbf{Exp}(\lambda \mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}). \tag{3.1}$$

We will also introduce the following lattice SHE. Let $\mathbf{Q}^{\mathsf{N}}$ be the solution to the SDE below for $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times \mathbb{X}_{\mathsf{N}}$ and with the same initial data as $\mathbf{Z}^{\mathsf{N}}$:

$$\mathrm{d}\mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^2(\alpha + \tfrac{1}{2}\lambda^2 \mathsf{N}^{-1}) \Delta \mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathrm{t} + \sqrt{2}\lambda \mathsf{N}^{\frac{1}{2}} \mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}}. \tag{3.2}$$

Above, we have used the lattice Laplacian notation $\Delta = -\nabla^{\mathbf{x}}_{-1} \nabla^{\mathbf{x}}_{1}$.

Proposition 3.1. *There exist $\zeta, \omega > 0$ independent of N such that with probability $1 - \mathrm{O}(\mathsf{N}^{-\zeta})$, we have*

$$\sup_{\mathrm{t} \in [0,1]} \max_{|\mathrm{x}| \leqslant 1} |\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\omega}. \tag{3.3}$$

Proof. The estimate (3.3) is the statement of Proposition 4.10 in [44] (recall that space is scaled by an extra factor of N in [44]). However, Proposition 4.10 in [44] is only shown with probability $1 - \mathrm{o}(1)$ as written. Fortunately, the proof of Proposition 4.10 in [44] is quantitative and is built on finitely many steps that each hold with probability $1 - \mathrm{O}(\mathsf{N}^{-\zeta})$ for some $\zeta > 0$. Indeed, the proof of Proposition 4.10 in [44] compares $\mathbf{Z}^{\mathsf{N}}$ to an auxiliary object $\mathbf{S}^{\mathsf{N}}$ (see (4.2) in [44]). Then $\mathbf{S}^{\mathsf{N}}$ is compared to $\mathbf{S}^{\mathsf{N},\zeta}$ (which is defined in (4.14) in [44] and uses a different parameter ζ than in this result), that is compared to $\mathbf{Q}^{\mathsf{N}}$. Moreover, each comparison holds deterministically on an intersection of $\mathrm{O}(1)$-many events of the form $\{\mathbf{e} \leqslant \mathsf{N}^{-\rho}\}$ for some error term $\mathbf{e}$ and fixed $\rho > 0$, whose probability is estimated by $\mathrm{O}(\mathsf{N}^{-\zeta})$ via Markov's or Chebyshev's inequality. See [44] for details. This completes the proof. □

Next, we will need to compare $\mathbf{Q}^{\mathsf{N}}$ to the following multiplicative-noise SHE for $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times \mathbb{R}$ and with initial data given by (3.1) at time $\mathrm{t} = 0$ (upon linearly interpolating $\mathbf{j}^{\mathsf{N}}$ from $\mathbb{X}_{\mathsf{N}}$ to $\mathbb{R}$):

$$\partial_{\mathrm{t}} \mathbf{Q}_{\mathrm{t},\mathrm{x}} = \alpha \partial^2_{\mathrm{x}} \mathbf{Q}_{\mathrm{t},\mathrm{x}} + \lambda \mathbf{Q}_{\mathrm{t},\mathrm{x}} \xi_{\mathrm{t},\mathrm{x}}. \tag{3.4}$$

The importance of SHE is that its logarithm solves the KPZ equation (1.2), in that it agrees with the solutions that we discussed in Section 1. To be precise, we have that $\mathbf{h} = \lambda^{-1} \log \mathbf{Q}$, where $\mathbf{h}$ solves (1.2) with initial data given by the linear interpolation extension of $\mathbf{j}^{\mathsf{N}}_{0,\cdot}$ from $\mathbb{X}_{\mathsf{N}}$ to $\mathbb{R}$; for the relationship between SHE and KPZ, see [18].

Lemma 3.2. *There exists a coupling of a standard two-sided Brownian motion to the initial data of $\mathbf{j}^{\mathsf{N}}$ (and, in particular, that of $\mathbf{Q}^{\mathsf{N}}$) such that for some $\zeta, \omega > 0$ independent of N, with probability $1 - \mathrm{O}(\mathsf{N}^{-\zeta})$, we have*

$$\sup_{\mathrm{t} \in [0,1]} \sup_{|\mathrm{x}| \leqslant 1} |\mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{Q}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\omega}. \tag{3.5}$$

The supremum above is over $\mathrm{x} \in \mathbb{R}$ with $|\mathrm{x}| \leqslant 1$, for which we extend $\mathbf{Q}^{\mathsf{N}}$ from $\mathbb{X}_{\mathsf{N}}$ to $\mathbb{R}$ via linear interpolation. (In particular, this makes it so that it does not matter if we take a supremum over $\mathrm{x} \in \mathbb{R}$ or $\mathrm{x} \in \mathbb{X}_{\mathsf{N}}$ above.)

Proof. The claim (3.5) without either supremum is essentially the content of Theorem 2.2 of [6]. (We note that [6] assumes a continuous regularization of SHE instead of a discrete one, but the proof carries over after cosmetic adjustments. Moreover, the initial data in [6] are assumed to be uniformly bounded in space, but the extension to initial data, such as two-sided Brownian motion, whose moment generating functions grow exponential-linearly in space is standard; see [7].) Deducing (3.5) as written follows from standard moment bounds on the space-time Hölder regularity of $\mathbf{Q}^{\mathsf{N}}, \mathbf{Q}$ (see (3.30) and (3.31) in [6], for instance). This completes the proof. □

Proposition 3.1 and Lemma 3.2 together yield a comparison estimate between $\mathbf{Z}^{\mathsf{N}}$ and $\mathbf{Q}$. In order to estimate the difference between their logarithms, we use the following positivity estimate on $\mathbf{Q}$; first, however, recall that $\mathbf{Q}$ has initial data $\mathbf{Exp}(\lambda \mathbf{j}^{\mathsf{N}}_{0,\cdot})$.

Lemma 3.3. *Fix any $\varepsilon > 0$ and $\mathrm{D} > 0$. With probability $1 - \mathrm{O}_{\varepsilon,\mathrm{D}}(\mathsf{N}^{-\mathrm{D}})$, we have*

$$\inf_{\mathrm{t} \in [0,1]} \inf_{|\mathrm{x}| \lesssim 1} |\mathbf{Q}_{\mathrm{t},\mathrm{x}}| \gtrsim \mathsf{N}^{-\varepsilon}. \tag{3.6}$$

Proof. By Corollary 4.8 in [23], we have the following negative moment estimate for any $p \geqslant 1$:

$$\sup_{t\in[0,1]} \sup_{|x|\lesssim 1} \mathbb{E}(|\mathbf{Q}_{t,x}|^{-2p}) \lesssim \mathbb{E}(|[e^{\alpha t\partial_x^2}\mathbf{Q}_{0,\cdot}]_x|^{-2p}).$$

We have the initial data estimate $\mathbb{E}(|\mathbf{Q}_{0,w}|^{-2p}) \lesssim_p \mathbf{Exp}(O_p(|w|))$, since $\mathbf{j}^{\mathsf{N}}_{0,w}$ is a sum of $\lesssim \mathsf{N}|w|$-many independent sub-Gaussian random variables of mean zero and variance parameter $\lesssim \mathsf{N}^{-1}$, while $\mathbf{j}^{\mathsf{N}}_{0,0} = 0$. (See Assumptions 2.3 and 2.4.) Moreover, by Jensen's inequality, we can move the negative moment on the RHS past the semigroup and onto $\mathbf{Q}_{0,\cdot}$. Thus, if $|x| \lesssim 1$ and $t \in [0,1]$, we obtain

$$\sup_{t\in[0,1]} \sup_{|x|\lesssim 1} \mathbb{E}(|\mathbf{Q}_{t,x}|^{-2p}) \lesssim_p 1.$$

Now, we apply (standard) high moment estimates for the space-time Hölder norms of $\mathbf{Q}$ (see (3.30) and (3.31) in [6]) to extend the previous pointwise estimate to

$$\mathbb{E}\Big(\sup_{t\in[0,1]} \sup_{|x|\lesssim 1} |\mathbf{Q}_{t,x}|^{-2p} \Big) \lesssim_p 1.$$

(To be more precise, we have $|\mathbf{Q}_{t,x}|^{-2p} - |\mathbf{Q}_{t,w}|^{-2p} \lesssim_p |\mathbf{Q}_{t,x} - \mathbf{Q}_{t,w}| \cdot (|\mathbf{Q}_{t,x}|^{-2p-1} + |\mathbf{Q}_{t,w}|^{-2p-1})$. Thus, if we use the Hölder inequality, we get $\mathbb{E}|(|\mathbf{Q}_{t,x}|^{-2p} - |\mathbf{Q}_{t,w}|^{-2p})|^{2q} \lesssim_{p,q} \mathbb{E}|\mathbf{Q}_{t,x} - \mathbf{Q}_{t,w}|^q \lesssim_q |x-w|^{q/2}$, where the last bound follows by the aforementioned moment estimates for $\mathbf{Q}$ as in [6]. This allows us to use the Kolmogorov continuity criterion to obtain moment estimates for $\|\mathbf{Q}_{t,\cdot}\|_{\mathscr{C}^{\upsilon}_{\mathrm{loc}}}$ for any fixed $\upsilon \in (0,1/2)$.) It now suffices to apply the Chebyshev inequality to complete the proof. □

Proposition 3.4. *There exist constants $\zeta, \kappa, \omega > 0$ such that with probability $1 - O(\mathsf{N}^{-\zeta})$,*

$$\sup_{t\in[0,1]} \sup_{|x|\leqslant \mathsf{N}^{\kappa}} |\mathbf{j}^{\mathsf{N}}_{t,x} - \mathbf{h}_{t,x}| \lesssim \mathsf{N}^{-\omega}. \tag{3.7}$$

Moreover, with probability $1 - O(\mathsf{N}^{-\zeta})$, we have

$$\sup_{x\in\mathbb{R}} \Big\{ (1+|x|)^{-1} |\mathbf{j}^{\mathsf{N}}_{1,x}| \Big\} + \sup_{x\in\mathbb{R}} \Big\{ (1+|x|)^{-1} |\mathbf{h}_{1,x}| \Big\} \lesssim \mathsf{N}^{\frac{1}{10}\kappa}. \tag{3.8}$$

(We clarify that there is nothing special about $1/10$ on the RHS; it is just small enough for future purposes.)

Proof. We start with (3.7). This estimate, if we restrict to $|x| \lesssim 1$ (instead of $|x| \leqslant \mathsf{N}^{\kappa}$) follows immediately with probability $1 - O(\mathsf{N}^{-\zeta})$ (for some $\zeta > 0$ fixed) by Proposition 3.1, Lemma 3.2, and Lemma 3.3.

We now claim that (3.7) holds if we instead restrict the spatial supremum to $|x - x_0| \lesssim 1$ for any fixed $x_0 \in \mathbb{R}$, again with probability $1 - O(\mathsf{N}^{-\zeta})$; that is, on such a high probability event, we have

$$\sup_{t\in[0,1]} \sup_{|x-x_0|\leqslant 1} |\mathbf{j}^{\mathsf{N}}_{t,x} - \mathbf{h}_{t,x}| \lesssim \mathsf{N}^{-\omega}.$$

To see this, we follow the reasoning given below:

- Recall that $\mathbf{h}$ is the solution to KPZ with initial data given by (an extension via linear-interpolation to $\mathbb{R}$ of) $\mathbf{j}^{\mathsf{N}}_{0,\cdot}$.
- Now, on the LHS of the previous display, let us subtract the quantity $\mathbf{j}^{\mathsf{N}}_{0,x_0}$ from both $\mathbf{j}^{\mathsf{N}}, \mathbf{h}$. (Since the supremum condition is defined by $\lesssim$ instead of $\leqslant$, we can assume that $x_0 \in \mathbb{X}_{\mathsf{N}}$.) We obtain the difference of solutions to (2.1) and (1.2) with initial data $\mathbf{j}^{\mathsf{N}}_{0,\cdot} - \mathbf{j}^{\mathsf{N}}_{0,x_0}$, because the evolutions (2.1) and (1.2) are equivariant with respect to constant shifts of the solution. Indeed, the RHS of (2.1) and (1.2) depend on the solution through gradients that vanish on constants.
- Finally, the space-time white noise and the initial data law $\mathbb{P}^0$ are invariant in law under space-time translations, so we can shift the LHS of the previous display in space to center around x_0 instead of the origin; indeed, we are only estimating the probability the above is true. In particular, we reduced to the case $x_0 = 0$. (We emphasize that the subtraction in the second bullet point yields Assumption 2.3 after the centering around x_0.)

Ultimately, we deduce that the previous display holds with probability $1 - O(\mathsf{N}^{-\zeta})$. Now, let us cover $[-\mathsf{N}^{\kappa}, \mathsf{N}^{\kappa}]$ with $O(\mathsf{N}^{\kappa})$-many intervals of length 2. By a union bound, we get (3.7) with probability $1 - O(\mathsf{N}^{\kappa}\mathsf{N}^{-\zeta})$, which is $1 - O(\mathsf{N}^{-\zeta})$ for a different value of $\zeta > 0$ if $\kappa = \zeta/2$, say.

We are left to show (3.8). To this end, as in Theorem 2.2 of [6], we know that all $p \geqslant 1$ and $\mathrm{x} \in \mathbb{R}$, we have

$$\mathbb{E}|\mathbf{Q}_{1,\mathrm{x}}|^{2p} \lesssim_p \mathbf{Exp}(\kappa_p|\mathrm{x}|).$$

for some $\kappa_p \lesssim_p 1$. At the same time, the negative moment estimates from the proof of Lemma 3.3 give

$$\mathbb{E}|\mathbf{Q}_{1,\mathrm{x}}|^{-2p} \lesssim_p \mathbf{e}^{\alpha \mathrm{t} \partial_{\mathrm{x}}^2}[\mathbb{E}|\mathbf{Q}_{0,\cdot}|^{-2p}]_{\mathrm{x}} \lesssim_p \mathbf{Exp}(\kappa_p|\mathrm{x}|)$$

for a possibly different $\kappa_p \lesssim 1$. Indeed, positive and negative moments of $\mathbf{Q}_{0,\cdot} = \mathbf{Exp}(\lambda \mathfrak{j}^{\mathsf{N}}_{0,\cdot})$ are bounded in the same way, i.e. using that $\mathrm{x} \mapsto \mathfrak{j}^{\mathsf{N}}_{0,\mathrm{x}}$ is a random walk of $\lesssim \mathsf{N}|\mathrm{x}|$-many independent sub-Gaussian and mean-zero increments of variance parameter $\lesssim \mathsf{N}^{-1}$ with pinning $\mathfrak{j}^{\mathsf{N}}_{0,0} = 0$. (This follows by Assumptions 2.2, 2.3, and 2.4.) Now, as in the proof of Lemma 3.3, moment estimates for the spatial Hölder regularity of SHE turn the previous two displays into the following, again for any $p \geqslant 1$ and a possibly different $\kappa_p \lesssim 1$:

$$\mathbb{E}\Big(\sup_{|\mathrm{w}-\mathrm{x}| \lesssim 1} |\mathbf{Q}_{1,\mathrm{w}}|^{2p} + \sup_{|\mathrm{w}-\mathrm{x}| \lesssim 1} |\mathbf{Q}_{1,\mathrm{w}}|^{-2p} \Big) \lesssim_p \mathbf{Exp}(\kappa_p|\mathrm{x}|).$$

Now, fix any integer $\ell \geqslant 1$. We can cover $[-\ell, \ell]$ by $\lesssim \ell$-many intervals of length 1 to deduce that

$$\mathbb{E}\Big(\sup_{|\mathrm{x}| \leqslant \ell} |\mathbf{Q}_{1,\mathrm{x}}|^2 \Big) + \mathbb{E}\Big(\sup_{|\mathrm{x}| \leqslant \ell} |\mathbf{Q}_{1,\mathrm{x}}|^{-2} \Big) \lesssim \ell \cdot \mathbf{Exp}(\mathrm{O}|\ell|).$$

Thus, by the Chebyshev inequality (with exponential moments), for any $\ell \geqslant 1$, we have

$$\begin{aligned} \mathbb{P}\Big(\sup_{|\mathrm{x}| \leqslant \ell} \ell^{-1}|\mathbf{h}_{1,\mathrm{x}}| \gtrsim \mathsf{N}^{\frac{1}{10}\kappa} \Big) &\lesssim \mathbf{Exp}(-2|\lambda|\ell \mathsf{N}^{\frac{1}{10}\kappa}) \cdot \Big\{ \mathbb{E}\Big(\sup_{|\mathrm{x}| \leqslant \ell} |\mathbf{Q}_{1,\mathrm{x}}|^2 \Big) + \mathbb{E}\Big(\sup_{|\mathrm{x}| \leqslant \ell} |\mathbf{Q}_{1,\mathrm{x}}|^{-2} \Big) \Big\} \\ &\lesssim \ell \cdot \mathbf{Exp}(-2|\lambda|\ell \mathsf{N}^{\frac{1}{10}\kappa} + \mathrm{O}(\ell)) \lesssim \mathbf{Exp}(-\mathsf{N}^{\frac{1}{20}\kappa}\ell). \end{aligned}$$

By a union bound, since the sum over $\ell \geqslant 1$ of the last term above is exponentially small in N, we deduce that

$$\mathbb{P}\Big(\sup_{\ell \geqslant 1} \sup_{|\mathrm{x}| \leqslant \ell} \ell^{-1}|\mathbf{h}_{1,\mathrm{x}}| \gtrsim \mathsf{N}^{\frac{1}{10}\kappa} \Big) \lesssim_{\mathrm{D}} \mathsf{N}^{-\mathrm{D}}$$

for any $\mathrm{D} > 0$. This gives the desired control on the second term on the LHS of (3.8).

Let us now control the first term on the LHS of (3.8). First, we note that with probability $1 - \mathrm{O}(\mathsf{N}^{-\zeta})$, we have $|\mathfrak{j}^{\mathsf{N}}_{1,0}| \lesssim \mathsf{N}^{\kappa/10}$. Indeed, this follows from the estimate in the previous paragraph for $\mathbf{h}$ combined with (3.7). Next, we note that it suffices to restrict the supremum to $\mathrm{x} \in \mathbb{X}_{\mathsf{N}}$ for the first term in (3.8), since the values of $\mathfrak{j}^{\mathsf{N}}$ off this lattice are linear interpolations of values on the lattice. Now, we recall that $\mathrm{x} \mapsto \mathfrak{j}^{\mathsf{N}}_{1,\mathrm{x}} - \mathfrak{j}^{\mathsf{N}}_{1,0}$ (for $\mathrm{x} \in \mathbb{X}_{\mathsf{N}} \cap [0, \infty)$) is a martingale with $\lesssim \mathsf{N}|\mathrm{x}|$-many mean-zero increments that are sub-Gaussian with variance parameter $\lesssim \mathsf{N}^{-1}$ by Assumption 2.2. (This is true at time 0, because $\phi_{0,\cdot} \sim \mathbb{P}^0$; since $\mathbb{P}^0$ is an invariant measure for the ϕ-process, it is true at time 1 as well.) Thus, $\mathfrak{j}^{\mathsf{N}}_{1,\mathrm{x}} - \mathfrak{j}^{\mathsf{N}}_{1,0}$ is sub-Gaussian of mean-zero and variance parameter $\lesssim \ell$. So, by the Doob maximal inequality and a Chernoff inequality for sums of independent sub-Gaussian random variables, we have the following estimate for any $\ell > 0$:

$$\begin{aligned} \mathbb{P}\Big(\sup_{\mathrm{x} \in [0,\ell]} (1 + |\ell|)^{-1}|\mathfrak{j}^{\mathsf{N}}_{1,\mathrm{x}} - \mathfrak{j}^{\mathsf{N}}_{1,0}| \geqslant \mathsf{N}^{\frac{1}{10}\kappa} \Big) &\lesssim \mathbf{Exp}(-\ell \cdot \mathsf{N}^{\frac{1}{10}\kappa}) \cdot \mathbb{E}(\mathbf{Exp}\{|\mathfrak{j}^{\mathsf{N}}_{1,\ell} - \mathfrak{j}^{\mathsf{N}}_{1,0}|\}) \\ &\lesssim \mathbf{Exp}(-\ell \cdot \mathsf{N}^{\frac{1}{10}\kappa} + \ell) \lesssim \mathbf{Exp}(-\ell \cdot \mathsf{N}^{\frac{1}{20}\kappa}). \end{aligned}$$

The same bound holds if we replace $[0, \ell]$ by $[-\ell, 0]$ as well by the same argument. Thus, by a union bound over all integers $\ell \geqslant 1$, we get the desired estimate for the first term on the LHS of (3.8), since the second line above sums over integers $\ell \geqslant 1$ to something that is exponentially small in N. This completes the proof. □

4. Proofs of Theorem 2.6 and Corollary 2.7

The technical goal of this section is to bootstrap the local-in-time estimate (3.7) to polynomial times. For this, we will need notation for the flows generated by the dynamics (1.2) and (2.1), respectively.

Notation 4.1. Given initial data $\mathbf{k}_\cdot : \mathbb{R} \to \mathbb{R}$ and times $0 \leqslant \mathrm{s} \leqslant \mathrm{t}$, let $\mathbf{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{KPZ}}[\mathbf{k}_\cdot] = \mathbf{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{KPZ}}[\mathbf{k}]$ denote the solution to (1.2) at time t with time-s data given by $\mathbf{k}$ (provided that (1.2) can be solved with this initial data).

Similarly, let $\mathbf{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{GL}}[\mathbf{k}_\cdot] = \mathbf{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{GL}}[\mathbf{k}]$ denote the solution to (2.1) at time t with time-s data given by the restriction of $\mathbf{k}$ from $\mathbb{R}$ to $\mathbb{X}_{\mathsf{N}}$. (If we are instead given initial data that is defined only on $\mathbb{X}_{\mathsf{N}}$, then we extend to $\mathbb{R}$ via linear interpolation when defining the action of $\mathbf{\Phi}^{\mathrm{s},\mathrm{t}}_{\mathrm{KPZ}}$.)

Finally, it will be convenient to define a similar solution map for SHE. For any $\mathbf{Y} : \mathbb{R} \to \mathbb{R}$ and $0 \leqslant s \leqslant t$, we let $\mathbf{\Phi}^{s,t}_{\mathrm{SHE}}[\mathbf{Y}]$ denote the solution to (3.4) at time t with data at time s given by $\mathbf{Y}$. (We may feed a function on $\mathbb{X}_{\mathsf{N}}$ into this solution map; the same extension via linear interpolation applies in this case.)

Remark 4.2. We clarify that in this section, if a function is a priori defined only on the lattice $\mathbb{X}_{\mathsf{N}} \subseteq \mathbb{R}$, we realize it as a function on $\mathbb{R}$ by extension-via-linear-interpolation. On the other hand, if a function is defined on $\mathbb{R}$, we realize it as a function on $\mathbb{X}_{\mathsf{N}}$ by restriction.

Our strategy runs as follows. First, we claim that for any $0 \leqslant s \leqslant r \leqslant t$, we have the identities

$$\mathbf{\Phi}^{s,t}_{\mathrm{KPZ}}[\mathbf{k}] = \mathbf{\Phi}^{r,t}_{\mathrm{KPZ}}[\mathbf{\Phi}^{s,r}_{\mathrm{KPZ}}[\mathbf{k}]] \quad \text{and} \quad \mathbf{\Phi}^{s,t}_{\mathrm{GL}}[\mathbf{k}] = \mathbf{\Phi}^{r,t}_{\mathrm{GL}}[\mathbf{\Phi}^{s,r}_{\mathrm{GL}}[\mathbf{k}]]$$

This follows by the flow-property of solutions to parabolic (S)PDEs and SDEs. Next, we build the following:

- Fix $\mathsf{t}_{\mathsf{N}} \in [0, \mathrm{O}(\mathsf{N}^{c})]$ as in Theorem 2.6, with $c > 0$ to be determined. Now, decompose $[0, \mathsf{t}_{\mathsf{N}}] = \mathbb{T}_1 \cup \ldots \cup \mathbb{T}_{\mathrm{K}_1}$, where $\mathrm{K} \lesssim \mathsf{N}^{c}$, and for $\mathrm{J} \in [\![1, \mathrm{K}-1]\!]$, we set $\mathbb{T}_{\mathrm{J}} = [\mathrm{J}-1, \mathrm{J}]$, where $\mathbb{T}_{\mathrm{K}} = [\mathrm{K}, \mathsf{t}_{\mathsf{N}}]$.
- For convenience, we will write $\mathbf{\Phi}^{\mathbb{T}_{\mathrm{J}}}$ to mean $\mathbf{\Phi}^{\mathrm{J}-1,\mathrm{J}}$.

Finally, we will let $\mathbf{h}$ denote the solution to (1.2) with initial data given by the extension-via-linear-interpolation of $\mathbf{j}^{\mathsf{N}}_{0,\cdot}$ from $\mathbb{X}_{\mathsf{N}}$ to $\mathbb{R}$. By the flow properties and the construction above, we can now write

$$\begin{aligned}
\mathbf{\Phi}^{0,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}] - \mathbf{\Phi}^{0,\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}] &= \mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{0,1}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]] - \mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{\Phi}^{0,1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]] \qquad (4.1)\\
&= \Big(\mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{0,1}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]] - \mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{0,1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]]\Big) + \Big(\mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{0,1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]] - \mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{\Phi}^{0,1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]]\Big)\\
&= \Big(\mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{0,1}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]] - \mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{0,1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]]\Big) + \Big(\mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{1,\cdot}] - \mathbf{\Phi}^{1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{1,\cdot}]\Big);
\end{aligned}$$

the second line follows by adding and subtracting the same term, and the third line follows since $\mathbf{j}^{\mathsf{N}}_{1,\cdot} = \mathbf{\Phi}^{0,1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]$ by construction. Continuing inductively, we arrive at the following.

Lemma 4.3. *We have the following deterministic formula:*

$$\mathbf{\Phi}^{0,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}] - \mathbf{\Phi}^{0,\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}] = \sum_{\mathrm{k} \in [\![0, \mathrm{K}-1]\!]} \Big(\mathbf{\Phi}^{\mathrm{k}+1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{\mathrm{k},\mathrm{k}+1}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}]] - \mathbf{\Phi}^{\mathrm{k}+1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{\mathrm{k},\mathrm{k}+1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}]]\Big). \qquad (4.2)$$

Proof. Take (4.1); for any $\mathrm{k} \geqslant 0$, by the same reasoning (but treating k as the initial time), we have

$$\mathbf{\Phi}^{\mathrm{k},\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}] - \mathbf{\Phi}^{\mathrm{k},\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}] = \Big(\mathbf{\Phi}^{\mathrm{k}+1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{\mathrm{k},\mathrm{k}+1}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}]] - \mathbf{\Phi}^{\mathrm{k}+1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{\Phi}^{\mathrm{k},\mathrm{k}+1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}]]\Big) + \Big(\mathbf{\Phi}^{\mathrm{k}+1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k}+1,\cdot}] - \mathbf{\Phi}^{\mathrm{k}+1,\mathsf{t}_{\mathsf{N}}}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k}+1,\cdot}]\Big).$$

Starting with $\mathrm{k} = 0$ as in (4.1) and applying this for each $\mathrm{k} = 0, \ldots, \mathrm{K}-1$ shows that that the LHS of (4.2) is the sum over such k of the first parenthetical on the RHS above, plus an additional term of the form $\mathbf{\Phi}^{\mathsf{t}_{\mathsf{N}},\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}} - \mathbf{\Phi}^{\mathsf{t}_{\mathsf{N}},\mathsf{t}_{\mathsf{N}}}_{\mathrm{SHE}}$, which vanishes since both maps are the identity. This completes the proof. □

Our goal is to estimate the summands on the RHS of (4.2). To start, we clarify the structure of each summand. The k-indexed term on the RHS of (4.2) is the difference of two solutions to the KPZ equation (1.2) (driven by the same noise) but with initial data $\mathbf{\Phi}^{\mathrm{k},\mathrm{k}+1}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}]$ and $\mathbf{\Phi}^{\mathrm{k},\mathrm{k}+1}_{\mathrm{GL}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{k},\cdot}]$, respectively. We emphasize that these initial data are *time*-1 *flows* under the (1.2) and (2.1) dynamics *with the same time*-k *data*. Thus, we can use Proposition 3.4 to estimate their difference. The missing ingredient to then estimate the difference between their images under the KPZ solution flow map is the following parabolic comparison principle for the KPZ equation.

Lemma 4.4. *First, we recall that* $\mathsf{t}_{\mathsf{N}} \leqslant \mathsf{N}^{c}$ *with* $c > 0$*, and fix any deterministic time* $s \in [0, \mathsf{t}_{\mathsf{N}}]$*. Now, suppose that* $\mathbf{k}^{(1)}, \mathbf{k}^{(2)} : \mathbb{X}_{\mathsf{N}} \to \mathbb{R}$ *are independent of the space-time white noise* $\xi_{u,w}$ *for* $u \geqslant s$*, and suppose that they satisfy*

$$\max_{|x| \leqslant \mathsf{N}^{\kappa}} |\mathbf{k}^{(1)}_x - \mathbf{k}^{(2)}_x| \lesssim \mathsf{N}^{-\omega} \quad \textit{and} \quad \max_{\ell=1,2} \sup_{x \in \mathbb{R}} \Big\{(1+|x|)^{-1} |\mathbf{k}^{(\ell)}_x|\Big\} \lesssim \mathsf{N}^{\frac{1}{10}\kappa}, \qquad (4.3)$$

where $\kappa, \omega > 0$ *are independent of* N*. If* $c > 0$ *is small enough (depending only on* κ, ω*), then for any* $D > 0$ *and any* $s \in [0, \mathsf{t}_{\mathsf{N}}]$*, we have the following estimate with probability* $1 - \mathrm{O}_D(\mathsf{N}^{-D})$*:*

$$\max_{|x| \lesssim \mathsf{N}^{c}} |\mathbf{\Phi}^{s,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{k}^{(1)}]_x - \mathbf{\Phi}^{s,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{k}^{(2)}]_x| \lesssim \mathsf{N}^{-\omega}. \qquad (4.4)$$

Proof. We have the identity $\mathbf{\Phi}^{s,t}_{KPZ}[\mathbf{k}^{(\ell)}] = \lambda^{-1}\log\mathbf{\Phi}^{s,t}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(\ell)})]$ for $\ell = 1, 2$, and thus (since $\lambda = \alpha^{-1}\beta \neq 0$ by assumption)

$$|\mathbf{\Phi}^{s,t_N}_{KPZ}[\mathbf{k}^{(1)}]_x - \mathbf{\Phi}^{s,t_N}_{KPZ}[\mathbf{k}^{(2)}]_x| \lesssim \Big|\log\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(1)})]_x - \log\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(2)})]_x\Big|.$$

Now, observe that (3.4) is linear in its solution. Thus, the solution map $\mathbf{\Phi}^{s,t_N}_{SHE}$ can be represented as integration against a kernel $\mathcal{Q}_{s,t_N}$. In particular, we have

$$\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(\ell)})]_x = \int_{\mathbb{R}}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell)}_y)dy,$$

where for any $(s,y)\in[0,\infty)\times\mathbb{R}$, the kernel $\mathcal{Q}_{s,t,x,y}$ satisfies SHE for all $(t,x)\in[s,\infty)\times\mathbb{R}$ with data $\mathcal{Q}_{s,s,x,y} = \delta_{x=y}$. By the positivity of solutions to SHE with probability measure initial data [30], we know that $\mathcal{Q}$ is a positive kernel. If we combine this with the first bound in (4.3) and a few elementary manipulations, we get

$$\begin{aligned}\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(1)})]_x &= \int_{|y|\leqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(1)}_y)dy + \int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(1)}_y)dy\\ &\leqslant \mathbf{Exp}\{O(N^{-\omega})\}\cdot\int_{|y|\leqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(2)}_y)dy + \int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(1)}_y)dy\\ &\leqslant \mathbf{Exp}\{O(N^{-\omega})\}\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(2)})]_x + \int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(1)}_y)dy.\end{aligned}$$

The same argument holds if we swap the roles of $(1),(2)$. Thus, for any $\ell_1,\ell_2\in\{1,2\}$, we have

$$\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)})]_x \leqslant \mathbf{Exp}\{O(N^{-\omega})\}\mathbf{\Phi}^{s,t_N}_{SHE}[\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_2)})]_x + \int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)}_y)dy. \quad (4.5)$$

We now claim that the following estimate holds for any $p\geqslant 1$, with explanation given afterwards, and in which $\mathbf{G}_{r,w}$ is the Gaussian heat kernel associated to $\partial_r - \alpha\partial_w^2$:

$$\mathbb{E}|\mathcal{Q}_{s,t,x,y}|^{2p} \lesssim_p e^{O_p(|t-s|)}\cdot|\mathbf{G}_{t-s,x-y}|^{2p}. \quad (4.6)$$

To see this, we use Proposition 2.3 in [6], which establishes (4.6) up to an exponentiated local-time-factor that is controlled by the exponential prefactor on the RHS above via Lemma 3.2 in [6]. (Even though Lemma 3.2 in [6] restricts to the setting $|t-s|\lesssim 1$, the exponentially-linear dependence on $|t-s|$ follows from the semigroup property for $\mathcal{Q}$, or by following the proof of Lemma 3.2 in [6], e.g. by chaining together estimates on length-$O(1)$ intervals covering $[s,t]$.) If we combine (4.6) and the second estimate in (4.3), we obtain the following estimate, in which we use $\|\cdot\|_{2p}^{2p} := \mathbb{E}_s|\cdot|^{2p}$, where the subscript indicates conditioning on data before time s:

$$\begin{aligned}\Big\|\int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)}_y)dy\Big\|_{2p} &\leqslant \int_{|y|\geqslant N^\kappa}\|\mathcal{Q}_{s,t_N,x,y}\|_{2p}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)}_y)dy\\ &\lesssim e^{O_p(t_N)}\int_{|y|\geqslant N^\kappa}\mathbf{G}_{t_N-s,x-y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)}_y)dy\\ &\lesssim e^{O_p(t_N)}\int_{|y|\geqslant N^\kappa}|t_N-s|^{-\frac12}\cdot e^{-\frac{\upsilon|x-y|^2}{t_N-s}}e^{\lambda N^{\frac{1}{10}\kappa}|y|}dy.\end{aligned}$$

Above, $\upsilon>0$ is fixed. If we restrict to $|x|\lesssim N^c$ for some $c>0$ small enough (but still independent of N), then we have $|x-y|\gtrsim|y|\gtrsim N^\kappa$. In particular, we obtain $\lambda\cdot N^{\kappa/10}\cdot|y| \ll |x-y|^2\cdot(t_N-s)^{-1}$ if $t_N\lesssim N^c$ as well. Thus, we can remove the second exponential factor in the last line at the cost of an upper bound if we change υ to $\frac12\upsilon$. A standard large deviations estimate for the Gaussian law shows that the remaining integral is $\lesssim\mathbf{Exp}(-\mu N^\kappa)$ with some $\mu>0$ fixed, again if $t_N,|x|\lesssim N^c$ for $c>0$ small enough but still independent of N. If we now combine this result with the Chebyshev inequality, then we obtain

$$\mathbb{P}\Big(\Big|\int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)}_y)dy\Big|\gtrsim\mathbf{Exp}(-N^{\frac12\kappa})\Big)\lesssim\mathbf{Exp}(O(N^{\frac12\kappa}))\cdot\mathbf{Exp}(-\mu N^\kappa)\lesssim\mathbf{Exp}(-\tfrac12 N^\kappa).$$

We now use a union bound to place a supremum over x in a discretization of $[-N^c,N^c]$ of mesh-length $\mathbf{Exp}(-N^{\kappa/2})$ at the cost of changing $\frac12 N^\kappa$ on the far RHS to $\frac13 N^\kappa$. Then, we apply a standard spatial Hölder-regularity estimate for SHE (see [6]) to show extend from said discretization of $[-N^c,N^c]$ to all of $[-N^c,N^c]$. Ultimately, we deduce that for any $D>0$, we have the following estimate with probability $1-O_D(N^{-D})$, in which $\delta>0$ is fixed:

$$\Big|\int_{|y|\geqslant N^\kappa}\mathcal{Q}_{s,t_N,x,y}\cdot\mathbf{Exp}(\lambda\mathbf{k}^{(\ell_1)}_y)dy\Big|\lesssim\mathbf{Exp}(-\delta N^\kappa).$$

Combining the previous display with (4.5), we obtain the following estimate, again for any $\ell_1, \ell_2 \in \{1,2\}$:

$$\mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell_1)})]_x \leqslant \mathbf{Exp}\{O(N^{-\omega})\}\mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell_2)})]_x + \mathbf{Exp}(-\delta N^{\kappa}) \tag{4.7}$$

Finally, we claim that for any $|x| \lesssim N^c$ and $\ell \in \{1,2\}$, we have the following with probability $1 - O_D(N^{-D})$:

$$\mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell)})]_x \gtrsim \mathbf{Exp}\{-N^{O(c)} - O(N^{\frac{1}{10}\kappa})\}. \tag{4.8}$$

Assume (4.8) for now. We obtain from (4.7)-(4.8) that $\mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell_1)})]_x \leqslant \mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell_2)})]_x(1+O(N^{-\omega}))$, again with probability $1 - O_D(N^{-D})$. Taking logarithms then gives $\mathbf{\Phi}^{s,t_N}_{\mathrm{KPZ}}[\mathbf{k}^{(\ell_1)}]_x \leqslant \mathbf{\Phi}^{s,t_N}_{\mathrm{KPZ}}[\mathbf{k}^{(\ell_2)}]_x + O(N^{-\omega})$. By using the same bound but swapping ℓ_1, ℓ_2, we obtain the desired estimate (4.4). Thus, it remains to prove (4.8). By Corollary 4.8 in [23][3], we have the negative moment estimate below for any $p \geqslant 1$:

$$\mathbb{E}|\mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell)})]_x|^{-2p} \lesssim_p \mathbf{Exp}(O_p(t_N^{O(1)})) \cdot |[\mathbf{G}_{t_N - s} * \mathbf{Exp}(\lambda \mathbf{k}^{(\ell)})]_x|^{-2p}.$$

By the second estimate in (4.3), if we restrict to $|x|, |t_N| \lesssim N^c$, the second factor on the RHS above is $\lesssim \mathbf{Exp}(N^{O(c)})$ by a standard integration estimate. (Intuitively, the Gaussian heat kernel has positive mass in a neighborhood of radius $O(N^c)$ of the origin, since $|x|, t_N \lesssim N^c$. On this set, we know that $|\mathbf{k}^{(\ell)}| \lesssim N^{\kappa/10} \cdot N^{O(c)}$ by (4.3).) Since we assumed that $t_N \lesssim N^c$, we ultimately deduce that

$$\sup_{|x| \lesssim N^c} \mathbb{E}|\mathbf{\Phi}^{s,t_N}_{\mathrm{SHE}}[\mathbf{Exp}(\lambda \mathbf{k}^{(\ell)})]_x|^{-2p} \lesssim_p \mathbf{Exp}\{O_p(N^{O(c)}) + O(N^{\frac{1}{10}\kappa})\}.$$

Another union bound argument (with spatial Hölder regularity of SHE solutions) shows that the desired estimate (4.8) holds with the desired probability $1 - O_D(N^{-D})$. This completes the proof. □

We are now in position to prove Theorem 2.6.

Proof of Theorem 2.6. First observe that the $\mathbf{\Phi}^{s,t}_{\mathrm{KPZ}}$ and $\mathbf{\Phi}^{s,t}_{\mathrm{GL}}$ operators are equivariant with respect to global shifts of the input function; that is, for any constant $a \in \mathbb{R}$, we get $\mathbf{\Phi}^{s,t}_{\mathrm{KPZ}}[\cdot + a] = \mathbf{\Phi}^{s,t}_{\mathrm{KPZ}}[\cdot] + a$ and $\mathbf{\Phi}^{s,t}_{\mathrm{GL}}[\cdot + a] = \mathbf{\Phi}^{s,t}_{\mathrm{GL}}[\cdot] + a$. Indeed, this follows since the RHS of (1.2) and (1.1) depend only on the solution through their (continuous and discrete) derivatives, which vanish on the constant $a \in \mathbb{R}$. Thus, we have (with explanation given afterwards)

$$\mathbf{\Phi}^{k,k+1}_{\mathrm{KPZ}}[\mathbf{j}^N_{k,\cdot}] - \mathbf{\Phi}^{k,k+1}_{\mathrm{GL}}[\mathbf{j}^N_{k,\cdot}] = \mathbf{\Phi}^{k,k+1}_{\mathrm{KPZ}}[\mathbf{m}^N_{k,\cdot}] - \mathbf{\Phi}^{k,k+1}_{\mathrm{GL}}[\mathbf{m}^N_{k,\cdot}],$$

where $\mathbf{m}^N_{k,\cdot} := \mathbf{j}^N_{k,\cdot} - \mathbf{j}^N_{k,0}$. Indeed, we have subtracted the same global shift from both solution maps on the LHS above. The advantage of this shift is that $\mathbf{m}^N_{k,\cdot}$ vanishes at $\cdot = 0$. Moreover, $-N^{1/2}\nabla^{\mathbf{x}}_{-1}\mathbf{m}^N_{k,\cdot} = -N^{1/2}\nabla^{\mathbf{x}}_{-1}\mathbf{j}^N_{k,\cdot} \sim \mathbb{P}^0$, because $\mathbb{P}^0$ is an invariant measure for the ϕ-process (see Section A.3 of [44]). Thus, by treating k as the initial time, we can use Proposition 3.4 to deduce that (4.3) is satisfied with the choices

$$\mathbf{k}^{(1)} = \mathbf{\Phi}^{k,k+1}_{\mathrm{KPZ}}[\mathbf{m}^N_{k,\cdot}] \quad \text{and} \quad \mathbf{k}^{(2)} = \mathbf{\Phi}^{k,k+1}_{\mathrm{GL}}[\mathbf{m}^N_{k,\cdot}],$$

at least with probability $1 - O(N^{-\zeta})$ for some fixed $\zeta > 0$. Thus, we can apply Lemma 4.4 to deduce

$$\max_{|x| \lesssim N^c} |\mathbf{\Phi}^{k+1,t_N}_{\mathrm{KPZ}}[\mathbf{\Phi}^{k,k+1}_{\mathrm{KPZ}}[\mathbf{j}^N_{k,\cdot}]]_x - \mathbf{\Phi}^{k+1,t_N}_{\mathrm{KPZ}}[\mathbf{\Phi}^{k,k+1}_{\mathrm{GL}}[\mathbf{j}^N_{k,\cdot}]]_x| \lesssim N^{-\omega}.$$

This holds with probability $1 - O(N^{-\zeta})$. A union bound shows that this holds for all $k \in [\![0, K-1]\!]$ simultaneously with probability $1 - O(KN^{-\zeta}) = 1 - O(N^c N^{-\zeta}) = 1 - O(N^{-\zeta/2})$ if $c > 0$ is small enough. Combining this with (4.2) yields the following with probability $1 - O(N^{-\zeta/2})$, again for $c > 0$ small enough but independent of N:

$$\max_{|x| \lesssim N^c} |\mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{j}^N_{0,\cdot}]_x - \mathbf{\Phi}^{0,t_N}_{\mathrm{GL}}[\mathbf{j}^N_{0,\cdot}]_x| \lesssim KN^{-\omega} \lesssim N^c N^{-\omega} \lesssim N^{-\frac{1}{2}\omega}. \tag{4.9}$$

By Notation 4.1, we have $\mathbf{\Phi}^{0,t_N}_{\mathrm{GL}}[\mathbf{j}^N_{0,\cdot}]_x = \mathbf{j}^N_{t_N,x}$ for all x. Moreover, recall from the statement of Theorem 2.6 and Notation 4.1 that $\mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{j}^N_{0,\cdot}]_x = \mathbf{h}^N_{t_N,x} + \mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{j}^N_{0,\cdot}]_x - \mathbf{h}^N_{t_N,x} = \mathbf{h}^N_{t_N,x} + (\mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{j}^N_{0,\cdot}]_x - \mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{w}_\cdot]_x)$ for a standard two-sided Brownian motion $\mathbf{w}_\cdot$. We now use the KMT coupling in [25] to couple $\mathbf{j}^N_{0,\cdot}, \mathbf{w}_\cdot$ so that

$$\max_{|x| \lesssim N^\kappa} |\mathbf{j}^N_{0,x} - \mathbf{w}_x| \lesssim N^{-\omega}$$

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with probability $1-\mathrm{O}(\mathsf{N}^{-\zeta})$, with some $\zeta,\omega,\kappa>0$ fixed. Moreover, we claim that with probability $1-\mathrm{O}(\mathsf{N}^{-\zeta})$, we have the following estimate (whose proof uses ideas from the argument for (3.8) that we explain for clarity):

$$\sup_{\mathrm{x}\in\mathbb{R}}\Big\{(1+|\mathrm{x}|)^{-1}|\mathbf{j}^{\mathsf{N}}_{0,\mathrm{x}}|\Big\}+\sup_{\mathrm{x}\in\mathbb{R}}\Big\{(1+|\mathrm{x}|)^{-1}|\mathbf{w}_{\mathrm{x}}|\Big\}\lesssim\log\mathsf{N}.$$

To show this, we decompose $\mathbb{R}=\cup_{\ell\geqslant 1}\mathbb{I}_{\ell}$, where $\mathbb{I}_{\ell}:=[-\ell,\ell]$ is the ball of radius ℓ. A standard application of the Doob maximal inequality for the martingale $\mathbf{w}_{\cdot}$ proves that $\|\mathbf{w}_{\cdot}\|_{\mathrm{L}^{\infty}(\mathbb{I}_{\ell})}$ is sub-Gaussian with variance parameter $\lesssim\ell$. So, $\mathbb{P}(\|\mathbf{w}_{\cdot}\|_{\mathrm{L}^{\infty}(\mathbb{I}_{\ell})}\gtrsim(\log\mathsf{N})\cdot\ell)\lesssim\mathbf{Exp}(-\upsilon(\log\mathsf{N})^{2}\ell)$ for some $\upsilon>0$ fixed. Summing this over $\ell\geqslant 1$ then controls the second term on the LHS of the display above. The same applies to the first term on the LHS as well, as $\mathbf{j}^{\mathsf{N}}_{0,\cdot}$ is a martingale (indexed by time-parameters in $\mathbb{X}_{\mathsf{N}}:=\mathsf{N}^{-1}\mathbb{Z}$) whose increments are sub-Gaussian of mean zero and variance parameter $\lesssim\mathsf{N}^{-1}$ because Assumption 2.2. The previous two displays and Lemma 4.4 yield

$$\max_{|\mathrm{x}|\lesssim\mathsf{N}^{\mathrm{c}}}|\mathbf{\Phi}^{0,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_{\mathrm{x}}-\mathbf{\Phi}^{0,\mathsf{t}_{\mathsf{N}}}_{\mathrm{KPZ}}[\mathbf{w}_{\cdot}]_{\mathrm{x}}|\lesssim\mathsf{N}^{-\omega};$$

we emphasize that this, as well as (4.9), holds with probability $1-\mathrm{O}(\mathsf{N}^{-\zeta})$ for some $\zeta>0$. As noted earlier, we can use this to get the desired estimate (2.8). This completes the proof. □

We conclude this section with a proof of Corollary 2.7.

Proof of Corollary 2.7. The desired statement is true for $\mathbf{h}$ in place of $\mathbf{j}^{\mathsf{N}}$ by [8]. Thus, it suffices to show that for some $\varepsilon>0$ fixed, we have the following estimate with probability $1-\mathrm{o}(1)$:

$$|\mathbf{j}^{\mathsf{N}}_{\mathsf{t}_{\mathsf{N}},0}-\mathbf{h}_{\mathsf{t}_{\mathsf{N}},0}|\lesssim\mathsf{N}^{-\varepsilon}\mathsf{t}_{\mathsf{N}}^{\frac{1}{3}}.$$

This, however, follows immediately by (2.8), so the proof is complete. □

5. Proofs of Theorem 2.8 and Corollary 2.9

We start with a proof of Theorem 2.8, as it happens to be fairly short.

Proof of Theorem 2.8. Let $\mathscr{B}$ be the "bad event" on which (4.9) fails. Let $\mathbb{P}^{\mathrm{stat}}$ denote the path-space probability measure induced by (2.1) with initial data that satisfies Assumption 2.4. We claim that $\mathbb{P}(\mathscr{B})^{2}\leqslant\mathsf{N}^{\gamma_{\mathrm{data}}}\mathbb{P}^{\mathrm{stat}}(\mathscr{B})$, where $\gamma_{\mathrm{data}}>0$ is from Assumption 2.5. This follows as the density of the law of the (2.1)-dynamics with initial data satisfying Assumption 2.5 with respect to that of the same dynamics with initial data satisfying Assumption 2.4 is equal to the density of the respective initial data, because the law of the dynamics is independent of the initial data. Thus, we can use the Cauchy-Schwarz inequality and (2.3) to change measure as claimed.

Theorem 2.6 (namely, the proof of (4.9)) gives $\mathbb{P}^{\mathrm{stat}}(\mathscr{B})\lesssim\mathsf{N}^{-\mathrm{c}}$ for some $\mathrm{c}>0$. Thus, if $\gamma_{\mathrm{data}}>0$ is small enough, the desired conclusion of Theorem 2.8 (for a possibly different $\mathrm{c}>0$) follows. □

Remark 5.1. We clarify that no comparison principle step is needed as in the end of the proof of Theorem 2.6, since by assumption in Theorem 2.8, the initial data for KPZ and (2.1) coincide.

We now move to the proof of Corollary 2.9. This starts with the following construction of initial data which approximates the narrow-wedge at a scale which is finer than the 1:2:3 KPZ scale, and which satisfies the initial data bound (2.3). The construction below resembles that in the proof of Theorem 2.7 in [44], which approximates a given initial data by vanishingly small $\mathbb{P}^{0}$ perturbations, though instead of being adapted to scales of order $\gg 1$, it is adapted to the time-scale t_{N} that we choose beforehand in the statement of Corollary 2.9.

Notation 5.2. Let $\mathbf{k}:\mathbb{X}_{\mathsf{N}}\to\mathbb{R}$ be a random function such that $\mathbf{k}_{0}=0$ and $\boldsymbol{\psi}\sim\mathbb{P}^{0}$, where $\boldsymbol{\psi}_{\mathrm{x}}:=-\mathsf{N}^{1/2}\nabla^{\mathbf{x}}_{-1}\mathbf{k}_{\mathrm{x}}$. Next, we set $\eta_{\mathsf{N}}:=(\log\log\mathsf{N})^{-1}$ and $\delta_{\mathsf{N}}:=(\log\log\log\mathsf{N})^{-1}$, and consider the "approximate wedge" function

$$\mathbf{k}^{\mathrm{wedge}}_{\mathrm{x}}:=-\mathrm{sgn}(\lambda)\cdot|\mathrm{x}|\cdot(\delta_{\mathsf{N}}\mathsf{t}_{\mathsf{N}}^{\frac{1}{3}})^{-1}.$$

(In words, the exponential $\mathbf{Exp}(\lambda\mathbf{k}^{\mathrm{wedge}})$, which gives the solution to SHE, is an approximation to the delta mass at the origin regularized at length-scale $\ll\mathsf{t}_{\mathsf{N}}^{2/3}$, with the exponent $1/3$ above coming from KPZ 1:2:3 scaling, at least up to a factor depending on $\lambda,\delta_{\mathsf{N}},\mathsf{t}_{\mathsf{N}}^{1/3}$.) Next, we construct the interval $\mathbb{I}_{\mathsf{N}}:=[-\eta_{\mathsf{N}}^{-1}\mathsf{t}_{\mathsf{N}}^{2/3},\eta_{\mathsf{N}}^{-1}\mathsf{t}_{\mathsf{N}}^{2/3}]\cap\mathbb{X}_{\mathsf{N}}$, as

well as the event

$$\mathscr{A}_{\text{wedge}} := \Big\{ \|\mathbf{k}_\cdot - \mathbf{k}_\cdot^{\text{wedge}}\|_{\mathrm{L}^\infty(\mathbb{I}_\mathsf{N})} \lesssim \eta_\mathsf{N} \mathsf{t}_\mathsf{N}^{\frac{1}{3}} \Big\}.$$

This is the event where $\mathbf{k}_\cdot, \mathbf{k}_\cdot^{\text{wedge}}$ are within the allowed error of size $\ll \mathsf{t}_\mathsf{N}^{1/3}$ as in Corollary 2.9. Lastly, let $\mathbb{P}_{\text{wedge}}$ denote the probability measure on $\mathbb{R}^{\mathbb{X}_\mathsf{N}}$ obtained by conditioning $\mathbb{P}^0$ on $\mathscr{A}_{\text{wedge}}$. (Observe $\mathscr{A}_{\text{wedge}}$ depends only on the gradients on $\mathbf{k}$, since $\mathbf{k}_0 = 0$ by construction. Thus, this probability measure $\mathbb{P}_{\text{wedge}}$ is well-defined.)

The first step towards Corollary 2.9 is checking that $\mathbb{P}_{\text{wedge}}$ satisfies the bound (2.3). In a nutshell, by diffusive scale-invariance of a standard Brownian motion, the event $\mathscr{A}_{\text{wedge}}$ is essentially asking for a Brownian motion to be within $\ll 1$ of a Lipschitz function of Lipschitz-norm $\gg 1$ on a domain of size $\gg 1$. By taking $\ll, \gg$ to mean vanishing and diverging slowly enough, Cameron-Marin theory for Brownian motion shows that the probability of such an event also vanishes slowly, and thus $\mathbb{P}_{\text{wedge}}$ is not "too far" from $\mathbb{P}^0$.

Lemma 5.3. *Let $\mathfrak{P}_{\text{wedge}}$ be the density defined by $\mathrm{d}\mathbb{P}_{\text{wedge}} = \mathfrak{P}_{\text{wedge}} \mathrm{d}\mathbb{P}^0$. We have*

$$\|\mathfrak{P}_{\text{wedge}}\|_{\mathrm{L}^2(\mathbb{P}^0)}^2 \lesssim \mathsf{N}^{\mathrm{o}(1)}. \tag{5.1}$$

Proof. First, by the definition of conditional probability, we have

$$\|\mathfrak{P}_{\text{wedge}}\|_{\mathrm{L}^2(\mathbb{P}^0)}^2 \lesssim \mathbb{P}(\mathscr{A}_{\text{wedge}})^{-1}.$$

Thus, it suffices to show that the probability on the RHS is $\gtrsim \mathsf{N}^{-\mathrm{o}(1)}$. By the KMT coupling in [25], it is enough to replace $\mathbf{k}_\cdot$ by a standard two-sided Brownian motion $\mathbf{w}$. (Indeed, $\mathbf{k}_\mathrm{x}$, as a process in $\mathrm{x} \geqslant 0$, is a random walk with $\lesssim \mathsf{N}|\mathrm{x}|$-many increments which are i.i.d. sub-Gaussian with zero mean, variance N, and variance parameter $\lesssim \mathsf{N}^{-1}$; see Assumption 2.2. The same is true for $\mathrm{x} \leqslant 0$ as well.)

By diffusive scale-invariance of the Brownian motion, we have

$$\mathbb{P}\Big(\|\mathbf{w}_\cdot\|_{\mathrm{L}^\infty(\mathbb{I}_\mathsf{N})} \lesssim \eta_\mathsf{N} \mathsf{t}_\mathsf{N}^{\frac{1}{3}}\Big) \gtrsim \mathbb{P}(\|\mathbf{w}_\cdot\|_{\mathrm{L}^\infty([-1,1])} \lesssim \eta_\mathsf{N}^{\mathrm{O}(1)}).$$

The reflection principle for Brownian motion shows that the probability on the RHS above is $\gtrsim \mathbf{Exp}(-\eta_\mathsf{N}^{-\mathrm{O}(1)})$, which is $\gtrsim \mathsf{N}^{-\mathrm{o}(1)}$ because $\eta_\mathsf{N} = (\log\log\mathsf{N})^{-1}$. (This is a standard small-ball probability estimate for the Wiener measure.) In particular, we have

$$\mathbb{P}\Big(\|\mathbf{w}_\cdot\|_{\mathrm{L}^\infty(\mathbb{I}_\mathsf{N})} \lesssim \eta_\mathsf{N} \mathsf{t}_\mathsf{N}^{\frac{1}{3}}\Big) \gtrsim \mathsf{N}^{-\mathrm{o}(1)}.$$

We now move to $\mathfrak{P}_{\text{wedge}}$. By the Girsanov theorem, the density for the law of $\mathbf{w}_\cdot$ with respect to that of $\mathbf{w}_\cdot - \mathbf{k}_\cdot^\wedge$ (both on $\mathbb{I}_\mathsf{N}$) has an L^2-norm (with respect to the law of $\mathbf{w}_\cdot - \mathbf{k}_\cdot^\wedge$) which is $\lesssim \mathbf{Exp}\{\mathrm{O}(\|\mathbf{k}_\cdot^{\text{wedge}}\|_{\mathrm{H}^1(\mathbb{I}_\mathsf{N})}^2)\}$; this is the standard Cameron-Martin factor. Because $\mathbf{k}^{\text{wedge}}$ has slopes $\lesssim \delta_\mathsf{N}^{-1}\mathsf{t}_\mathsf{N}^{-1/3}$, we get $\|\mathbf{k}_\cdot^{\text{wedge}}\|_{\mathrm{H}^1(\mathbb{I}_\mathsf{N})}^2 \lesssim \delta_\mathsf{N}^{-2}\mathsf{t}_\mathsf{N}^{-2/3}|\mathbb{I}_\mathsf{N}| \lesssim \delta_\mathsf{N}^{-2}\eta_\mathsf{N}^{-1}$. Thus, by the Cauchy-Schwarz inequality, we obtain

$$\mathbb{P}(\|\mathbf{w}_\cdot\|_{\mathrm{L}^\infty(\mathbb{I}_\mathsf{N})} \lesssim \eta_\mathsf{N}\mathsf{t}_\mathsf{N}^{\frac{1}{3}})^2 \lesssim \mathbf{Exp}\{\mathrm{O}(\delta_\mathsf{N}^{-2}\eta_\mathsf{N}^{-1})\} \cdot \mathbb{P}(\|\mathbf{w}_\cdot - \mathbf{k}_\cdot^{\text{wedge}}\|_{\mathrm{L}^\infty(\mathbb{I}_\mathsf{N})} \lesssim \eta_\mathsf{N}\mathsf{t}_\mathsf{N}^{\frac{1}{3}}).$$

We showed earlier in this proof that the LHS is $\gtrsim \mathsf{N}^{-\mathrm{o}(1)}$. Since $\eta_\mathsf{N}^{-1} = \log\log\mathsf{N}$ and $\delta_\mathsf{N}^{-2} = (\log\log\log\mathsf{N})^2$, from this, we obtain that the probability on the RHS above is also $\gtrsim \mathsf{N}^{-\mathrm{o}(1)}$. This completes the proof. □

We are now in position to prove Corollary 2.9.

Proof of Corollary 2.9. Throughout this proof, we sample $\mathbf{j}_{0,\cdot}^\mathsf{N}$ with the rules $\mathbf{j}_{0,0}^\mathsf{N} = 0$ and $\phi_{0,\cdot} \sim \mathbb{P}_{\text{wedge}}$. By [4], we know that for some constants $\mathrm{c}_1, \mathrm{c}_2 \in \mathbb{R}$ with $\mathrm{c}_2 \neq 0$, we have (in the flow language of Notation 4.1)

$$(\mathrm{c}_2 \mathsf{t}_\mathsf{N}^{\frac{1}{3}})^{-1} \cdot (\mathbf{\Phi}_{\mathrm{KPZ}}^{0,\mathsf{t}_\mathsf{N}}[\mathbf{f}^{\mathrm{nw}}]_0 - \mathrm{c}_1\mathsf{t}_\mathsf{N}) \Rightarrow \mathrm{TW}_2,$$

in which $\mathbf{\Phi}_{\mathrm{KPZ}}^{0,\mathsf{t}_\mathsf{N}}[\mathbf{f}^{\mathrm{nw}}]_0 = \lambda^{-1}\log\mathbf{\Phi}_{\mathrm{SHE}}^{0,\mathsf{t}_\mathsf{N}}[\delta_0]$, so that $\mathbf{f}_\cdot^{\mathrm{nw}} := \lambda^{-1}\log\delta_0(\cdot)$ formally. Thus, by Theorem 2.8, it suffices to show that for some sequence $\varepsilon_\mathsf{N} \to 0$, with probability $1 - \mathrm{o}(1)$, we have

$$|\mathbf{\Phi}_{\mathrm{KPZ}}^{0,\mathsf{t}_\mathsf{N}}[\mathbf{j}_{0,\cdot}^\mathsf{N}]_0 - \mathbf{\Phi}_{\mathrm{KPZ}}^{0,\mathsf{t}_\mathsf{N}}[\mathbf{f}^{\mathrm{nw}}]_0| \lesssim \varepsilon_\mathsf{N}\mathsf{t}_\mathsf{N}^{\frac{1}{3}}, \tag{5.2}$$

Before we present the argument, let us explain the main idea, since the proof requires careful attention to various exponents (that are ultimately related to 1:2:3 KPZ scaling).

- First, we will take the exponential of $\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0$ to obtain a solution to the SHE in (3.4). Because the SHE is linear in its solution, we can express $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0)$ as an integral of its initial data against the fundamental solution of SHE, which itself has the law of a narrow-wedge solution to SHE.
- Second, we will apply spatial decay estimates for the fundamental solution of SHE, which are adapted to 1:2:3 scaling of KPZ, to establish that at time t_N, the contribution of the initial data of $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0)$ outside the set $\mathbb{I}_\mathsf{N}$ from Notation 5.2 is negligible, namely $\mathbf{Exp}(\mathrm{o}(\mathsf{t}_\mathsf{N}^{1/3}))$. (Recall that $\mathbb{R} \setminus \mathbb{I}_\mathsf{N}$ only contains points at distance $\gg \mathsf{t}_\mathsf{N}^{2/3}$ from the origin; we emphasize the relationship between the spatial exponent $2/3$, the time exponent 1, and the fluctuation exponent $1/3$ to provide intuition for why such a result should be true.)
- The previous bullet point reduces the estimation of $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0)$ to the integral of its initial data against the fundamental solution of the SHE on the set $\mathbb{I}_\mathsf{N}$. After this, use that $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0)$ decays exponentially quickly outside an interval of the form $[-\rho_\mathsf{N} \mathsf{t}_\mathsf{N}^{2/3}, \rho_\mathsf{N} \mathsf{t}_\mathsf{N}^{2/3}]$, for $\rho_\mathsf{N} \ll 1$, again up to an ultimately negligible factor of the form $\mathbf{Exp}(\mathrm{o}(\mathsf{t}_\mathsf{N}^{1/3}))$. Then, we use that the fundamental solution of SHE at time t_N has regularity at scale $\mathsf{t}_\mathsf{N}^{1/3}$, so on $[-\rho_\mathsf{N} \mathsf{t}_\mathsf{N}^{2/3}, \rho_\mathsf{N} \mathsf{t}_\mathsf{N}^{2/3}]$, it is almost constant (and equal to its value at the origin, namely the narrow-wedge solution to SHE at the origin time t_N). Once again, this is all up to a negligible factor $\mathbf{Exp}(\mathrm{o}(\mathsf{t}_\mathsf{N}^{1/3}))$.
- The last step shows that $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0) \approx \mathbf{Exp}(\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0) \cdot \mathbf{Exp}(\mathrm{o}(\mathsf{t}_\mathsf{N}^{1/3}))$; taking logarithms completes the argument.
- Finally, before we proceed, we emphasize the importance of estimates for the fundamental solution of the SHE that are stable under 1:2:3 KPZ scaling. Fortunately, these estimates are all available because of the work [11], which we will cite often.

Let us proceed with the argument. We first use

$$\mathrm{e}^{\lambda \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{KPZ}}[\mathbf{j}^{\mathsf{N}}_{0,\cdot}]_0} = \mathbf{\Phi}^{0,\mathsf{t}_\mathsf{N}}_{\mathrm{SHE}}[\mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\cdot}}]_0 = \int_{\mathbb{R}} \mathcal{Q}_{0,\mathsf{t}_\mathsf{N},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y},$$

where $\mathcal{Q}$ is the fundamental solution to the SHE (3.4). In particular, it has the law of the narrow-wedge solution to (1.2) at time t_N. (As a function of (s,y), the function $\mathcal{Q}_{\mathsf{t}_\mathsf{N}-\mathrm{s},\mathsf{t}_\mathsf{N},0,\mathrm{y}}$ satisfies the adjoint equation corresponding to (3.4) by standard parabolic theory with initial data $\delta_{\mathrm{y}=0}$ and $\mathrm{s}=0$. But the operator on the RHS of (3.4), which acts on the solution therein, is self-adjoint with respect to Lebesgue measure. Thus, $\mathcal{Q}_{\mathsf{t}_\mathsf{N}-\mathrm{s},\mathsf{t}_\mathsf{N},0,\mathrm{y}}$ is a solution of SHE itself in (s,y)-variables with a time-reversed space-time white noise, which has the same law as space-time white noise itself.) So, by Lemma 4.1 in [11], for any $\varepsilon > 0$, in N is large enough, then with probability $1-\varepsilon$,

$$\mathcal{Q}_{0,\mathsf{t}_\mathsf{N},0,\mathrm{y}} \lesssim \mathbf{Exp}(\mathrm{a}_1 \mathsf{t}_\mathsf{N} + \mathrm{a}_2 \mathsf{t}_\mathsf{N}^{\frac{1}{3}} - \mathrm{a}_3 \tfrac{\mathrm{y}^2}{\mathsf{t}_\mathsf{N}}) \quad \text{for all } \mathrm{y} \in \mathbb{R} \tag{5.3}$$

for some deterministic constants $\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3 \lesssim 1$ with $\mathrm{a}_3 \gtrsim 1$. (We note that [11] assumes that $\alpha, \beta = \frac{1}{2}$ in (1.2), but by rescaling the equation (1.2) in space-time and solution space, we can always reduce to $\alpha, \beta = \frac{1}{2}$.) This gives

$$\begin{aligned}\int_{\mathbb{R}} \mathcal{Q}_{0,\mathsf{t}_\mathsf{N},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} &= \int_{\mathbb{I}_\mathsf{N}} \mathcal{Q}_{0,\mathsf{t}_\mathsf{N},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} + \int_{\mathbb{R}\setminus\mathbb{I}_\mathsf{N}} \mathcal{Q}_{0,\mathsf{t}_\mathsf{N},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} \\ &= \int_{\mathbb{I}_\mathsf{N}} \mathcal{Q}_{0,\mathsf{t}_\mathsf{N},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} + \mathrm{O}\Big(\int_{\mathbb{R}\setminus\mathbb{I}_\mathsf{N}} \mathbf{Exp}(\mathrm{a}_1 \mathsf{t}_\mathsf{N} + \mathrm{a}_2 \mathsf{t}_\mathsf{N}^{\frac{1}{3}} - \mathrm{a}_3 \tfrac{\mathrm{y}^2}{\mathsf{t}_\mathsf{N}}) \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} \Big).\end{aligned}$$

Now, we fix a constant $\mathrm{m}_\mathsf{N} \gg 1$ that diverges slowly and small $\nu > 0$ (both to be determined). We claim that with probability $1 - \mathrm{o}(1)$, we have the following for some $\upsilon > 0$ fixed (as long as $\mathrm{m}_\mathsf{N} \gg 1$ slowly enough)

$$\mathbf{1}_{\mathrm{y} \notin \mathbb{I}_\mathsf{N}} \cdot \lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}} \leqslant \mathrm{m}_\mathsf{N} \mathsf{t}_\mathsf{N}^{\frac{1}{3}} + \nu \mathsf{t}_\mathsf{N}^{-1} \mathrm{y}^2 - \upsilon \eta_\mathsf{N}^{-1} \delta_\mathsf{N}^{-1} \mathsf{t}_\mathsf{N}^{\frac{1}{3}} \leqslant \nu \mathsf{t}_\mathsf{N}^{-1} \mathrm{y}^2 - \tfrac{1}{2} \upsilon \eta_\mathsf{N}^{-1} \delta_\mathsf{N}^{-1} \mathsf{t}_\mathsf{N}^{\frac{1}{3}}.$$

We justify the previous display as follows:

(1) The second inequality follows since $\mathrm{m}_\mathsf{N} \to \infty$ slowly, while $\eta_\mathsf{N}, \delta_\mathsf{N} \to 0$. So, we are left with the first bound.

(2) By construction in Notation 5.2, we have the following (with notation explained after):

$$\mathbf{1}_{\mathrm{y} \notin \mathbb{I}_\mathsf{N}} \cdot \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}} = \mathbf{1}_{\mathrm{y} > \sup \mathbb{I}_\mathsf{N}} (\mathbf{j}^{\mathsf{N}}_{0,\sup \mathbb{I}_\mathsf{N}} + \mathbf{j}^{\mathsf{N},>}_{0,\mathrm{y}}) + \mathbf{1}_{\mathrm{y} < \inf \mathbb{I}_\mathsf{N}} (\mathbf{j}^{\mathsf{N}}_{0,\inf \mathbb{I}_\mathsf{N}} + \mathbf{j}^{\mathsf{N},<}_{0,\mathrm{y}}).$$

We start with $\mathbf{j}^{\mathsf{N},>}$. As a process $\mathrm{y} \in \mathbf{j}^{\mathsf{N},>}_{0,\mathrm{y}}$ in $\mathrm{y} \in \mathbb{X}_\mathsf{N}$ such that $\mathrm{y} > \sup \mathbb{I}_\mathsf{N}$, is a random walk with independent increments that are mean-zero and sub-Gaussian with variance parameter $\lesssim \mathsf{N}^{-1}$ by Assumption 2.2. Indeed, in Notation 5.2, we do not condition on the behavior of $\mathbf{j}^{\mathsf{N}}_{0,\cdot}$ outside the set $\mathbb{I}_\mathsf{N}$, so the gradients $-\mathsf{N}^{1/2} \nabla^{\mathbf{X}}_{-1} \mathbf{j}^{\mathsf{N}}_{0,\cdot}$ outside $\mathbb{I}_\mathsf{N}$ are distributed via $\mathbb{P}^0$. By the same token, the process $\mathrm{y} \mapsto \mathbf{j}^{\mathsf{N},<}_{0,\mathrm{y}}$, with $\mathrm{y} \in \mathbb{X}_\mathsf{N}$ such

that $\mathrm{y} < \inf \mathbb{I}_{\mathsf{N}}$, is a random walk with independent increments that are mean-zero and sub-Gaussian with variance parameter $\lesssim \mathsf{N}^{-1}$ (with the time-variable y parameterized in the backwards-time-direction).

(3) Again by construction in Notation 5.2, we have that $\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}} = -|\lambda| \cdot |\mathrm{y}| \cdot \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{-1/3} + \mathrm{O}(\eta_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{1/3})$ if $\mathrm{y} \in \mathbb{I}_{\mathsf{N}}$; this is deterministic. If we now choose $\mathrm{y} = \sup \mathbb{I}_{\mathsf{N}}, \inf \mathbb{I}_{\mathsf{N}}$ and recall that $\eta_{\mathsf{N}}, \delta_{\mathsf{N}} \to 0$ while $|\lambda| > 0$ is fixed, then

$$\mathbf{j}^{\mathsf{N}}_{0,\sup \mathbb{I}_{\mathsf{N}}}, \mathbf{j}^{\mathsf{N}}_{0,\inf \mathbb{I}_{\mathsf{N}}} \leqslant -|\lambda| \cdot \eta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{\frac{2}{3}} \cdot \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{-\frac{1}{3}} + \mathrm{O}(\eta_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{1/3}) \leqslant -\upsilon \eta_{\mathsf{N}}^{-1} \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}$$

Thus, it suffices to show that with probability $1 - \mathrm{o}(1)$, we have

$$\mathbf{1}_{\mathrm{y} > \sup \mathbb{I}_{\mathsf{N}}} \mathbf{j}^{\mathsf{N},>}_{0,\mathrm{y}} + \mathbf{1}_{\mathrm{y} < \inf \mathbb{I}_{\mathsf{N}}} \mathbf{j}^{\mathsf{N},<}_{0,\mathrm{y}} \leqslant \mathrm{m}_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}} + \nu \mathrm{t}_{\mathsf{N}}^{-1} \mathrm{y}^2.$$

We give the argument for the first term on the LHS; the argument for the second term therein is the same. We recall that for any $\mathrm{y} > \sup \mathbb{I}_{\mathsf{N}}$, the value $\mathbf{j}^{\mathsf{N},>0}_{0,\mathrm{y}}$ is the sum of $\lesssim \mathsf{N}|\mathrm{y}|$-many independent mean-zero increments that are sub-Gaussian with variance parameter $\lesssim \mathsf{N}^{-1}$, again by Assumption 2.2. (Recall $\mathbf{j}^{\mathsf{N},>}$ was a process on $\mathbb{X}_{\mathsf{N}} := \mathsf{N}^{-1}\mathbb{Z}$, so any unit interval has $\lesssim \mathsf{N}$-many increments.) Thus, by the KMT coupling, it is enough to replace $\mathbf{j}^{\mathsf{N},>}$ by a standard Brownian motion $\mathbf{q}_{\cdot}$ on $[\sup \mathbb{I}_{\mathsf{N}}, \infty)$ that is pinned so that $\mathbf{q}_{\sup \mathbb{I}_{\mathsf{N}}} = 0$. Let us now assume $\sup \mathbb{I}_{\mathsf{N}} = 0$, mostly out of notational convenience. (This reduction can also be justified upon shifting the time-parameter in $\mathbf{q}_{\cdot}$ by $-\sup \mathbb{I}_{\mathsf{N}}$.) We now cite Lemma 2.12 in [11]. This shows that for any $\nu > 0$, there is a constant $\nu_0 > 0$ such that for all $\sigma \geqslant 1$, we have

$$\mathbb{P}\Big(\sup_{\mathrm{y} \in [0,\infty)} (\sigma + \nu|\mathrm{y}|^2)^{-1} \mathbf{q}_{\mathrm{y}} \geqslant 1 \Big) \lesssim_{\nu} \mathrm{e}^{-\nu_0 \sigma^{\frac{3}{2}}}.$$

By diffusive scale-invariance of Brownian motion, we can replace $\mathbf{q}_{\mathrm{y}} \mapsto \mathrm{t}_{\mathsf{N}}^{-1/3} \mathbf{q}_{\mathrm{t}_{\mathsf{N}}^{2/3}}$, so that

$$\mathbb{P}\Big(\sup_{\mathrm{y} \in [0,\infty)} (\sigma + \nu|\mathrm{y}|^2)^{-1} \mathrm{t}_{\mathsf{N}}^{-\frac{1}{3}} \mathbf{q}_{\mathrm{t}_{\mathsf{N}}^{2/3} \mathrm{y}} \geqslant 1 \Big) = \mathbb{P}\Big(\sup_{\mathrm{y} \in [0,\infty)} (\mathrm{t}_{\mathsf{N}}^{\frac{1}{3}} \sigma + \mathrm{t}_{\mathsf{N}}^{-1} \nu |\mathrm{y}|^2)^{-1} \mathbf{q}_{\mathrm{y}} \geqslant 1 \Big) \lesssim_{\nu} \mathrm{e}^{-\nu_0 \sigma^{\frac{3}{2}}}.$$

(The identity follows by moving the prefactor $\mathrm{t}_{\mathsf{N}}^{-1/3}$ inside the factor $(\sigma + \nu|\mathrm{y}|^2)^{-}$ and then re-parameterizing $\mathrm{y} \mapsto \mathrm{t}_{\mathsf{N}}^{-2/3} \mathrm{y}$.) We now choose $\sigma = \mathrm{m}_{\mathsf{N}} \to \infty$ (slowly) to conclude the desired estimate.

Therefore, we get the following (for $\nu > 0$ small), in which we absorb $\mathrm{a}_2 \mathrm{t}_{\mathsf{N}}^{1/3}$ by exponents of the form $\gg \mathrm{t}_{\mathsf{N}}^{1/3}$:

$$\begin{aligned}
&\int_{\mathbb{R} \setminus \mathbb{I}_{\mathsf{N}}} \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} + \mathrm{a}_2 \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}} - \mathrm{a}_3 \tfrac{\mathrm{y}^2}{\mathrm{t}_{\mathsf{N}}}) \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} \\
&\lesssim \int_{\mathbb{R} \setminus \mathbb{I}_{\mathsf{N}}} \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} + \mathrm{a}_2 \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}} - \mathrm{a}_3 \tfrac{\mathrm{y}^2}{\mathrm{t}_{\mathsf{N}}}) \mathbf{Exp}(-\upsilon \eta_{\mathsf{N}}^{-1} \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}) \mathbf{Exp}(\nu \tfrac{\mathrm{y}^2}{\mathrm{t}_{\mathsf{N}}}) \mathrm{d}\mathrm{y} \\
&\lesssim \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} - \tfrac{1}{2} \upsilon \eta_{\mathsf{N}}^{-1} \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}) \int_{\mathbb{R} \setminus \mathbb{I}_{\mathsf{N}}} \mathbf{Exp}(-\tfrac{\mathrm{a}_3 \mathrm{y}^2}{2\mathrm{t}_{\mathsf{N}}}) \mathrm{d}\mathrm{y} \\
&\lesssim \mathrm{t}_{\mathsf{N}} \cdot \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} - \tfrac{1}{2} \upsilon \eta_{\mathsf{N}}^{-1} \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}) \lesssim \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} - \tfrac{1}{3} \upsilon \eta_{\mathsf{N}}^{-1} \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}).
\end{aligned}$$

We move to the integration over $\mathbb{I}_{\mathsf{N}}$. First, we set $\mathbb{L}_{\mathsf{N}} := [-\rho_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{2/3}, \rho_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{2/3}]$ for $\rho_{\mathsf{N}} \to 0$ slowly. Then, we write

$$\int_{\mathbb{I}_{\mathsf{N}}} \mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} = \int_{\mathbb{L}_{\mathsf{N}}} \mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} + \int_{\mathbb{I}_{\mathsf{N}} \setminus \mathbb{L}_{\mathsf{N}}} \mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y}.$$

Let us take the second term on the RHS. On $\mathbb{I}_{\mathsf{N}}$, we have (as noted before) $\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}} = -|\lambda| \cdot |\mathrm{y}| \cdot \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{-1/3} + \mathrm{O}(\eta_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{1/3})$ on $\mathbb{I}_{\mathsf{N}}$, by construction in Notation 5.2. Thus, if $\mathrm{y} \in \mathbb{I}_{\mathsf{N}} \setminus \mathbb{L}_{\mathsf{N}}$, we have $\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}} \leqslant -|\lambda| \cdot \rho_{\mathsf{N}} \delta_{\mathsf{N}}^{-1} \mathrm{t}_{\mathsf{N}}^{1/3} + \mathrm{O}(\eta_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{1/3})$. For $\rho_{\mathsf{N}} \to 0$ slowly enough, we have $|\lambda| \rho_{\mathsf{N}} \delta_{\mathsf{N}}^{-1} \geqslant \delta_{\mathsf{N}}^{-1/2}$. Thus, if we use (5.3), then the second term on the RHS is

$$\begin{aligned}
\int_{\mathbb{I}_{\mathsf{N}} \setminus \mathbb{L}_{\mathsf{N}}} \mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,\mathrm{y}} \mathrm{e}^{\lambda \mathbf{j}^{\mathsf{N}}_{0,\mathrm{y}}} \mathrm{d}\mathrm{y} &\lesssim \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} + \mathrm{a}_2 \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}) \cdot \mathbf{Exp}(-\delta_{\mathsf{N}}^{-\frac{1}{2}} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}} + \mathrm{O}(\eta_{\mathsf{N}} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}})) \cdot |\mathbb{I}_{\mathsf{N}}| \\
&\lesssim \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} - \delta_{\mathsf{N}}^{-\frac{1}{3}} \mathrm{t}_{\mathsf{N}}^{\frac{1}{3}}).
\end{aligned}$$

with probability $\geqslant 1 - \varepsilon$, where $\varepsilon > 0$ is small. We now take the integral over $\mathbb{L}_{\mathsf{N}}$. We claim that for $\mathrm{y} \in \mathbb{L}_{\mathsf{N}}$, with probability $1 - \mathrm{o}(1)$, we have $\mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,\mathrm{y}} = \mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,0} \mathbf{Exp}(\mathrm{o}(\mathrm{t}_{\mathsf{N}}^{1/3}))$. To this end, we use Theorem 1.2 in [11], so that

$$\mathcal{Q}_{0,\mathrm{t}_{\mathsf{N}},0,\mathrm{y}} = \mathbf{Exp}(\mathrm{a}_1 \mathrm{t}_{\mathsf{N}} + \mathrm{t}_{\mathsf{N}}^{1/3} \boldsymbol{\mathcal{H}}_{\mathrm{t}_{\mathsf{N}}^{-2/3} \mathrm{y}}),$$

in which $\boldsymbol{\mathcal{H}}_\cdot$ is absolutely continuous with respect to Wiener measure and is thus $\mathscr{C}^{1/3}$ with probability 1. Thus, with probability $1 - o(1)$, we can replace the $\boldsymbol{\mathcal{H}}$-term above by its value at 0, if we pay a cost of $t_N^{1/3} m_N t_N^{-2/3} \cdot |y|$ in the exponential, where $m_N \to \infty$ slowly as before. This cost is $\lesssim m_N \rho_N t_N^{1/3} \ll t_N^{1/3}$ if $m_N \to \infty$ slowly enough. Therefore, if we again use $\lambda \mathbf{j}^N_{0,y} = -|\lambda| \cdot |y| \cdot \delta_N^{-1} t_N^{-1/3} + O(\eta_N t_N^{1/3})$ on $\mathbb{I}_N$, we get

$$\int_{\mathbb{L}_N} \mathcal{Q}_{0,t_N,0,y} e^{\lambda \mathbf{j}^N_{0,y}} dy = \mathcal{Q}_{0,t_N,0,0} \int_{\mathbb{L}_N} \mathbf{Exp}\{o(t_N^{\frac{1}{3}})\} e^{\lambda \mathbf{j}^N_{0,y}} dy = \mathcal{Q}_{0,t_N,0,0} \cdot \mathbf{Exp}\{o(t_N^{\frac{1}{3}})\}.$$

Ultimately, we obtain (with probability $\geqslant 1 - \varepsilon$ for $\varepsilon > 0$ small) that

$$\int_{\mathbb{I}_N} \mathcal{Q}_{0,t_N,0,y} e^{\lambda \mathbf{j}^N_{0,y}} dy = \mathcal{Q}_{0,t_N,0,0} \cdot \mathbf{Exp}\{o(t_N^{\frac{1}{3}})\} + O(\mathbf{Exp}(a_1 t_N - \delta_N^{-\frac{1}{3}} t_N^{\frac{1}{3}})).$$

Combining this with the estimate for the integral outside $\mathbb{I}_N$ as well as (5.3) (at $y = 0$) gives

$$\begin{aligned}\int_{\mathbb{R}} \mathcal{Q}_{0,t_N,0,y} e^{\lambda \mathbf{j}^N_{0,y}} dy &= \mathcal{Q}_{0,t_N,0,0} \cdot \mathbf{Exp}\{o(t_N^{\frac{1}{3}})\} + O(\mathbf{Exp}(a_1 t_N - \delta_N^{-\frac{1}{3}} t_N^{\frac{1}{3}})) + \mathbf{Exp}(a_1 t_N - \tfrac{1}{3} \upsilon \eta_N^{-1} \delta_N^{-1} t_N^{\frac{1}{3}}) \\ &= \mathcal{Q}_{0,t_N,0,0} \cdot \Big(\mathbf{Exp}\{o(t_N^{\frac{1}{3}})\} + O\{\mathbf{Exp}(-\omega_N t_N^{\frac{1}{3}})\} \Big),\end{aligned}$$

where $\omega_N \to \infty$. Now, recall that the far LHS above is $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{j}^N_{0,\cdot}]_0)$. Moreover, the factor $\mathcal{Q}_{0,t_N,0,0}$ is the solution to SHE (3.4) with initial data δ_0; thus, it is equal to $\mathbf{Exp}(\lambda \mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{f}^{\mathrm{nw}}]_0)$. If we take logarithms, we get

$$\begin{aligned}|\lambda| \cdot |\mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{j}^N_{0,\cdot}]_0 - \mathbf{\Phi}^{0,t_N}_{\mathrm{KPZ}}[\mathbf{f}^{\mathrm{nw}}]_0| &= \Big| \log \Big(\mathbf{Exp}\{o(t_N^{\frac{1}{3}})\} + O\{\mathbf{Exp}(-\omega_N t_N^{\frac{1}{3}})\} \Big) \Big| \\ &= \Big| \log \Big(\mathbf{Exp}\{o(t_N^{\frac{1}{3}})\} \Big\{ 1 + O\{\mathbf{Exp}(-\tfrac{1}{2} \omega_N t_N^{\frac{1}{3}})\} \Big\} \Big) \Big| \\ &\lesssim o(t_N^{\frac{1}{3}}) + \mathbf{Exp}(-\tfrac{1}{2} \omega_N t_N^{\frac{1}{3}}) \lesssim o(t_N^{\frac{1}{3}}).\end{aligned}$$

Since $\lambda \neq 0$ is fixed, we can divide by $|\lambda|$ to obtain the desired estimate (5.2). This completes the proof. □

REFERENCES

[1] A. Aggarwal, I. Corwin, M. Hegde, "Scaling limit of the colored ASEP and stochastic six-vertex models", arXiv:2403.01341, 2024.
[2] A. Aggarwal, I. Corwin, M. Hegde, "KPZ fixed point convergence of the ASEP and stochastic six-vertex models", arXiv:2412.18117, 2024.
[3] A. Aggarwal, J. Huang, "Strong characterization for the Airy line ensemble", *Inventiones Mathematicae*, 243, 293-605, 2026.
[4] G. Amir, I. Corwin, J. Quastel, "Probability distribution of the free energy of the continuum directed polymer model in (1+1)-dimensions", *Communications on Pure and Applied Mathematics*, 64, 466-537, 2011.
[5] J. Baik, E. Rains, "Limiting distributions for a polynuclear growth model with external sources", *Journal of Statistical Physics*, 100, 523-541, 2000.
[6] L. Bertini, N. Cancrini, "The stochastic heat equation: Feynman-Kac formula and intermittence", *Journal of Statistical Physics*, 78, 1377-1401, 1995.
[7] L. Bertini, G. Giacomin, "Stochastic Burgers and KPZ Equations from Particle Systems", *Communications in Mathematical Physics*, 183, 3, 571-606, 1997.
[8] A. Borodin, I. Corwin, P. Ferrari, B. Vetö, "Height fluctuations for the stationary KPZ equation", *Mathematical Physics, Analysis, and Geometry*, 18, 20, 2015.
[9] A. Chandra, L. Ferdinand, "A flow approach to the generalized KPZ equation", arXiv:2402.03101, 2024.
[10] I. Corwin, A. Hammond, "Brownian Gibbs property for Airy line ensembles", *Inventiones Mathematicae*, 195, 441-508, 2014.
[11] I. Corwin, A. Hammond, "The KPZ line ensemble", *Probability Theory and Related Fields*, 166, 1-2, 67-185, 2016.
[12] I. Corwin, J. Quastel, "Crossover distributions at the edge of the rarefaction fan", *Annals of Probability*, 41, 3A, 1243-1314, 2013.
[13] D. Dauvergne, J. Ortmann, B. Virag, "The directed landscape", *Acta Mathematica*, 229, 2, 201-285, 2022.
[14] J. Diehl, M. Gubinelli, N. Perkowski, "The Kardar-Parisi-Zhang equation as scaling limit of weakly asymmetric interacting Brownian motions", *Communications in Mathematical Physics*, 354, 549-589, 2017.
[15] S. Ganguly, A. Hammond, "Stability and chaos in dynamical last passage percolation", *Communications of the American Mathematical Society*, 4, 387-479, 2024.
[16] M. Gubinelli, N. Perkowski, "The Hairer-Quastel universality result at stationarity", *RIMS Symposium on Stochastic Analysis on Large Scale Interacting System*, B59, 101-115, 2016.
[17] M. Gubinelli, N. Perkowski, "KPZ reloaded", *Communications in Mathematical Physics*, 349, 165-269, 2017.
[18] M. Hairer, "Solving the KPZ equation", *Annals of Mathematics*, 178, 2, 559-664, 2013.
[19] M. Hairer, J. Quastel, "A class of growth models rescaling to KPZ", *Forum of Mathematics, Pi*, 6, E3, 2018.
[20] M. Hairer, H. Shen, "A central limit theorem for the KPZ equation", *Annals of Probability*, 45, 4167-4221, 2017.
[21] M. Hairer, W. Xu. "Large scale limit of interface fluctuation models", *Annals of Probability*, 47, 6, 3478-3550, 2019.
[22] P. C. Hohenberg, B. I. Halperin, "Theory of dynamic critical phenomena", *Reviews of Modern Physics*, 49, 435, 1977.

[23] Y. He, K. Lê, "Asymptotics of the density of parabolic Anderson random fields", *Annales de l'Institut Henri Poincaré*, 58, 1, 105-133, 2022.
[24] M. Kardar, G. Parisi, Y.-C. Zhang, "Dynamic scaling of growing interfaces", *Physical Review Letters* 56, 889, 1986.
[25] J. Komlós, P. Major, G. Tusnády, "An approximation of partial sums of independent rv's and the sample df. I", *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 32, 111-131, 1975.
[26] F. Kong, H. Wang, W. Xu, "Hairer-Quastel universality for KPZ – polynomial smoothing mechanisms, general nonlinearities and Poisson noise", arXiv:2403.06191, 2024.
[27] F. Kong, W. Zhao, "A frequency-independent bound on trigonometric polynomials of Gaussians and applications", *Journal of Functional Analysis*, 288, 3, 110705, 2025.
[28] A. Kupiainen, M. Marcozzi, "Renormalization of generalized KPZ equation", *Journal of Statistical Physics*, 166, 876-902, 2017.
[29] K. Matetski, J. Quastel, D. Remenik, "The KPZ fixed point", *Acta Mathematica*, 227, 1, 115-203, 2021.
[30] C. Mueller, "On the support of solutions to the heat equation with noise", *Stochastics and Stochastic Reports*, 37, 4, 225-245, 1991.
[31] C. Noack, B. Landon, P. Sosoe, "KPZ-type fluctuation exponents for interacting diffusions in equilibrium", *Annals of Probability*, 51, 3, 1138-1191, 2023.
[32] B. Landon, P. Sosoe, "Upper tail bounds for stationary KPZ models", *Communications in Mathematical Physics*, 401, 2, 1311-1335, 2023.
[33] B. Landon, P. Sosoe, "Tail bounds for the O'Connell-Yor polymer", *Electronic Journal of Probability*, 1-47, 2024.
[34] B. Landon, P. Sosoe, "Tail estimates for the stationary stochastic six vertex model and ASEP", *Probability and Mathematical Physics*, 6, 1327-1378, 2025.
[35] N. Perkowski, T. Rosati, "The KPZ equation on the real line", *Electronic Journal of Probability*, 24, 1-56, 2019.
[36] J. Quastel, "Introduction to KPZ", *Current Developments in Mathematics*, 2011, 1, 2011.
[37] T. Sasamoto, H. Spohn, "Exact height distributions for the KPZ equation with narrow wedge initial condition", *Nuclear Physics B*, 834, 523-542, 2010.
[38] T. Sasamoto, H. Spohn, "One-dimensional Kardar-Parisi-Zhang equation: an exact solution and its universality", *Physical Review Letters*, 104, 230602, 2010.
[39] T. Sasamoto, H. Spohn, "The crossover regime for the weakly asymmetric simple exclusion process", *Journal of Statistical Physics*, 140, 209-231, 2010.
[40] H. Spohn, *Large Scale Dynamics of Interacting Particles*, Springer-Verlag Berlin Heidelberg, 1 edition, 1991.
[41] X. Wu, "KPZ Convergence of the KPZ line ensemble", *International Mathematics Research Notices*, 2023, 22, 18901-18957, 2023.
[42] K. Yang, "Hairer-Quastel universality in non-stationarity via energy solution theory", *Electronic Journal of Probability*, 28, 1-26, 2023.
[43] K. Yang, "Time-inhomogeneous KPZ equation from non-equilibrium Ginzburg-Landau SDEs", *Electronic Journal of Probability*, 30, 1-155, 2025.
[44] K. Yang, "KPZ equation from a class of nonlinear SPDEs in infinite volume", *Probability Theory and Related Fields*, 2026.
[45] X. Zhang, R. Zhu, X. Zhu, "Singular HJB equations with applications to KPZ on the real line", *Probability Theory and Related Fields*, 183, 789-869, 2022.